

Option-Implied Risk Premia and Cryptocurrency Market Regimes

Caio Almeida* Maria Grith[†] Ratmir Miftachov[‡] Zijin Wang[§]

Abstract

We analyze the first- and second-moment risk premia in the Bitcoin market using options and realized returns. The results show that Bitcoin is much more volatile and has higher premia than the S&P 500 stock index. By decomposing the return premium across different regions of the return state space, we find that while most of the S&P 500 equity premium comes from mildly negative returns, the corresponding negative Bitcoin returns (between three and one standard deviation from the mean) account for only one-third of the total Bitcoin premium (BP). The decomposition of the Bitcoin variance risk premium (BVRP) reveals a dominant positive contribution from upside return states, in contrast to the S&P 500, where the VRP is primarily driven by downside returns. Further, by applying a novel clustering algorithm to a collection of estimated Bitcoin option-implied risk-neutral densities, we find that risk premia vary over time across two distinct market volatility regimes. The low-volatility regime is associated with lower BP and BVRP, and the share of the total premia attributable to positive returns is higher in magnitude. These results suggest that in stable markets, Bitcoin investors place relatively higher importance on upside return and variance risk.

Keywords: Bitcoin, Options, Return Risk Premium, Variance Risk Premium, Pricing Kernel, Cryptocurrency, Density Clustering, Nonparametric and Semiparametric Estimation

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*E-mail: calmeida@princeton.edu. Department of Economics, Princeton University, USA

[†]E-mail: maria.grith@neoma-bs.fr. Finance Department, Neoma Business School, France

[‡]E-mail: ratmir.miftachov@hu-berlin.de. School of Business and Economics; Institute of Mathematics, Humboldt-Universität zu Berlin, Germany

[§]E-mail: wangzijin_900@bankcomm.com. Postdoctoral Programme of Bank of Communications, Bank of Communications, Shanghai, China

1 Introduction

The cryptocurrency market plays a central role in the digital economy, comprising thousands of cryptocurrencies, numerous exchanges, and a global market capitalization of billions of dollars. As this market expands, financial derivatives linked to cryptocurrencies and crypto-traded funds are gaining popularity. Similar to traditional assets, crypto derivatives can provide essential insight into risk premia, reflecting investors’ compensation for taking on risk. Although there is extensive literature on risk premia for traditional assets, particularly equities, there is limited analysis regarding these premia in the cryptocurrency market. This paper aims to fill this gap by conducting a comprehensive analysis of the Bitcoin risk premia using options data, with a focus on the Bitcoin return premium (BP) and the variance risk premium (BVRP).

We start by compiling a reliable joint dataset of Bitcoin returns and option prices to document several key stylized features of the unconditional Bitcoin risk premia. Then, we propose a novel two-stage approach to identify the variations in Bitcoin risk premia and analyze their response to market conditions. In the first stage, we construct a series of risk-neutral density (RND) functions implied by option prices that, jointly with the returns series, enable us to assess the properties of risk premia. In the second stage, we utilize functional data analysis classification tools to classify RNDs into homogeneous clusters that characterize different market regimes. For these regimes, we calculate the corresponding conditional risk premia (BP and BVRP).

Our analysis focuses on Bitcoin – the first decentralized and most widely adopted cryptocurrency. We view Bitcoin as a digital asset and utilize traditional asset pricing tools to derive insights into its behavior.¹ Many previous studies have explored the factors that influence Bitcoin prices to assess risk premia. In this analysis, however, we specifically focus on the risk premia implied by Bitcoin returns and option prices from the Deribit trading platform – the largest exchange for crypto-derivatives globally.²

¹In the U.S., Bitcoin falls under the Commodity Futures Trading Commission (CFTC). Some studies compare Bitcoin with commodities (Alexander and Imeraj, 2021; Bianchi, 2020; Hou et al., 2020), while others treat it as a currency (see Cao and Celik, 2021; Schilling and Uhlig, 2019; Uhlig, 2024).

²We also examine risk premia for Ethereum as a robustness check. The results are available in Section A.10 of the Appendix.

Popular approaches use option prices and index returns to estimate and decompose risk premia as a function of index returns, with minimal assumptions on the data-generating process. These methods rely on general no-arbitrage conditions, enabling data-driven estimation, and at the same time, can be linked to standard macro-finance models (Beason and Schreindorfer, 2022, Schreindorfer and Sichert, 2025). Our analysis is also based on a theoretical setting derived under a representative-agent setting, in which the stochastic discount factor is tied to the market portfolio as a proxy for the agent’s wealth (Martin, 2017, Chabi-Yo and Loudis, 2020, Chabi-Yo and Loudis, 2024). In terms of preferences and wealth allocation, applying these approaches separately to Bitcoin implicitly treats this market in isolation and abstracts from a setting in which investors jointly allocate across asset classes (e.g., in the S&P 500 index alongside Bitcoin). In principle, our results may capture interactions between markets. However, existing empirical evidence suggests that the Bitcoin market largely moves independently of traditional macro-financial markets during our sample period, so we expect such interactions to be limited.³

Institutional and structural differences between cryptocurrency and equity markets imply distinct patterns in option-implied risk premia, documented in the literature as follows. Bitcoin exhibits substantially higher volatility and risk premia than the S&P 500, reflecting compensation for risk in a speculative and less mature market (Liu and Tsyvinski, 2021). Cryptocurrency returns appear to be driven largely by crypto-specific risk factors, indicating that the observed returns compensate investors for exposure to distinct sources of risk (Liu, Tsyvinski, and Wu, 2022). Unlike equities, where premia are primarily associated with downside risk, Bitcoin premia receive a significant contribution from positive return states (Scaillet, Treccani, and Trevisan, 2020). Earlier studies find that, in contrast to the traditional assets, the term structure of Bitcoin risk premia is upward sloping, particularly during bull markets (Foley et al., 2022). In this article, we re-examine the magnitude and return-space decomposition of BP and BVRP, as well as their variation across market regimes and maturities, and dependence on exogenous variables.

³Bianchi (2020) reports limited correlations between cryptocurrencies and traditional asset classes, with macro indicators having minimal impact on crypto markets. Liu and Tsyvinski (2021) finds no significant link between Bitcoin returns and consumption or production growth. Alexander and Imeraj (2021) note that, before the COVID-19 pandemic, Bitcoin’s VRP did not align with that of other assets; however, it became highly correlated with the VRP of equity and gold during the pandemic. Chen et al. (2021) show that Bitcoin’s pricing kernel is decoupled from the consumption kernel, with minimal impact from real-economy shocks, like the Covid-19 crisis.

Employing semi- and nonparametric statistical techniques on options data and return time series enables us to identify patterns in the Bitcoin market with minimal restrictions and compare them with those of traditional financial assets. We start by calculating unconditional measures of risk premia. For a one-month horizon, we find that the BP averages 66% per annum, while the BVRP averages 14%, both of which are significantly higher than the premia of traditional assets. For comparison, the average lower bound of S&P 500 equity premium (EP) for a one-month horizon is 5% (Martin, 2017). Additionally, focusing on the second moment risk premium, Bollerslev, Tauchen, and Zhou (2009) find a 2% variance risk premium (VRP) for the S&P 500. Although Bitcoin demonstrates significantly greater volatility compared to traditional assets like the S&P 500—by a considerable margin—its annualized Sharpe Ratio (SR) is approximately 0.8,⁴ which is nearly twice that of the S&P 500.⁵

Although the unconditional BP can be estimated using Bitcoin returns and interest rates alone, understanding how it varies across different return states necessitates a more comprehensive dataset, which option prices can provide. Using the equity premium (EP) decomposition method proposed by Beason and Schreindorfer (2022), we find that large positive returns (ranging from 20% to 60%⁶) account for 39% of the Bitcoin premium. In contrast, less than one-third of the premium for the S&P 500 is attributable to positive returns. We find that while most of the S&P 500 equity premium comes from mildly negative returns, the corresponding negative Bitcoin returns between three and one standard deviations account for only one-third of the total Bitcoin premium (BP). To verify the stability of our results across model specifications, we also employ series-expansion-based estimators following Chabi-Yo and Loudis (2020), Chabi-Yo and Loudis (2024), and Schreindorfer and Sichert (2025). While under these specifications, the BP decomposition shows that most of the BP is attributable to negative returns, the share of BP attributable to positive returns is significantly higher than that of S&P 500. These findings suggest that the right side of the Bitcoin return distribution contains crucial information about the Bitcoin market.

⁴This result is consistent with the SR of about 1.07 implied by monthly returns in Liu and Tsyvinski (2021).

⁵The SR for the S&P 500 is approximately 0.5 (Dew-Becker, Giglio, and Kelly, 2021), and the SR for the SPY is 0.45 per annum (Feng and He, 2022). Sections B.4 and B.5 in the Appendix provide a more detailed comparison between Bitcoin and traditional markets.

⁶Bitcoin returns from 20% to 60% correspond to approximately one to three standard deviations of the return distribution.

A natural follow-up question is whether risk premia change with market conditions. To explore this, we use information derived solely from option prices to identify the main Bitcoin market regimes. The data from single-day Bitcoin prices can be unreliable due to a low signal-to-noise ratio. In contrast, Bitcoin options are available for various strikes and maturities each day, supplying more reliable information about market pricing. Additionally, their forward-looking nature makes options more sensitive to expected market conditions, making them ideal for capturing signals regarding changes in the market. We propose a novel clustering algorithm for a collection of RNDs estimated from option prices. The objective is to classify them into clusters, allowing the mean density within a cluster to reflect stylized information for asset pricing in a specific market regime. We identify two main clusters: a high-volatility (HV) regime and a low-volatility (LV) regime. In the high-volatility regime, the contribution of large positive and negative returns to the BP is roughly the same, around 30%, for the selected return segments. On the other hand, under the low-volatility regime, around 52% of the BP is attributable to large positive returns between 20% and 60%. These results suggest Bitcoin investors are more concerned about upside risk in the calmer market. Furthermore, all measures we consider indicate that the BVRP is lower in the LV regime. Our findings are robust across alternative maturities and across cryptocurrencies. Extending the analysis to 9- and 45-day horizons, we find that the term structure of BP is downward sloping for the BP and upward sloping for the BVRP.

To gain further insight into the distribution of risk premia across return states, we rely on the decomposition framework of Chabi-Yo and Loudis (2024), decomposing both the BP and BVRP into three regions: down, center, and up. This approach allows us to additionally examine how the BVRP varies across different regions of the return distribution. In contrast to evidence for the S&P 500, where the left tail dominates the VRP, we find economically significant contributions from both tails. In particular, the downside region contributes negatively to the BVRP—consistent with a discount for downside exposure—while the upside region contributes positively, reflecting compensation for upside variance risk.

Finally, we examine the drivers of the total risk premia and their return-state components. We find predictability only for BP and its components. The predictability is primarily driven by

upside return states and is closely linked to sentiment, particularly that of bullish and individual investors, and investor attention. We also find that the upside return premium is higher following favorable economic conditions.

This paper is structured as follows. We start with a brief literature review in the next subsection. Section 2 provides a detailed overview of the data. Sections 3 and 4 introduce the theoretical framework and the estimation methodology, followed by Section 5, which highlights the main findings. Section 6 provides a thorough summary of the findings and implications of this article, and offers recommendations for future research.

1.1 Related Literature

In this article, we empirically document stylized features of risk premia in the Bitcoin market. We estimate unconditional risk premia implied by options data and returns (BP and BVRP), decompose BP and BVRP on return states, propose a statistically-based clustering method, and track how risk premia vary within these clusters and whether they are predictable. Our work connects with four main strands of literature.

The first strand analyzes risk premia in cryptocurrencies using Bitcoin options and returns. While studies on cryptocurrency indexes are at a more developed stage, research about derivatives of digital assets is still in its early stages, with limited exploration of cryptocurrency options-implied risk premia. We contribute to this literature by documenting the stylized features and variation of BP and BVRP across return states and market regimes using an extended dataset. Our unconditional risk premia estimates align with those reported in the existing literature on Bitcoin; i.e., the return premium in Foley et al. (2022) and Wilson (2024), and the variance premium in Alexander and Imeraj (2021)⁷ and Winkel and Härdle (2023). Focusing on the time-series properties of Bitcoin variance risk premium, Alexander and Imeraj (2021) find that it spikes before large positive or negative returns on Bitcoin. The atypical behavior of Bitcoin is documented by Hou et al. (2020) as an "inverse leverage effect", resembling commodities, which

⁷Alexander and Imeraj (2021) are the first to construct a Bitcoin VIX Index and estimate VRP for Bitcoin, using options traded on Deribit exchange.

suggests that positive changes in the prices are associated with higher volatility. In a similar vein, but employing a different approach, we notice a positive relationship between returns and volatility. Alexander and Imeraj (2023) document an upward-sloping implied volatility curve, particularly during bullish phases of Bitcoin’s price. We find that the positive right slope of Bitcoin options implied volatility is steeper during the low-volatility regime.

The second strand of literature focuses on the decomposition of risk premia on return states. The risk price is naturally expressed as a function of returns through the pricing kernel (PK), and numerous studies have characterized it using options and returns data (e.g., Aït-Sahalia and Lo, 2000; Jackwerth, 2000), with a primary focus on the equity market. We will discuss this literature in detail in what follows. In contrast, the literature on the decomposition of EP on return states is relatively sparse, focusing primarily on equity premia. We contribute to this literature by documenting the first and second moment premia decomposition and PK patterns for the Bitcoin market. Our pricing kernel unconditional estimate for a one-month horizon is hump-shaped, featuring an increasing segment for small and medium negative returns of the Bitcoin index. For the Bitcoin market, Winkel and Härdle (2023) explore the PK term structure and find that empirical PKs are W-shaped for longer-dated options. Our paper employs the EP decomposition of Beason and Schreindorfer (2022) using refined non-parametric techniques and documents the differences between the S&P 500 and Bitcoin markets. Relying on the same methodology, Almeida, Freire, and Hizmeri (2024) find evidence for the prominent contribution of right-tail return states to the EP and VRP of the S&P 500 index for options with ultra-short maturities. A complementary approach for studying risk premia for different return segments is proposed by Chabi-Yo and Loudis (2024), who find that EP and VRP of S&P 500 index are largely driven by the left tail of the return distribution. We also implement this approach and show that most of the BP and BVRP is attributable to positive returns.

The third strand of the literature examines the conditional estimation of option-implied risk measures using clustering methods for functional data. Identifying the relevant factors for conditioning is particularly important in emerging markets like Bitcoin, which lack a clear understanding of the risks that contribute to the overall risk premia. Previous research on traditional

assets has focused on how different market mechanisms and factors affect various risk measures. We review this literature in the next paragraph. By contrast, our paper introduces a data-driven approach to identify the drivers and estimate conditional premia that is both economically and statistically sound. We specifically investigate the information found in option prices to reveal relevant information for conditioning. We estimate nonparametrically the risk-neutral density (RND) implied by options in liquid markets (Breen and Litzenberger, 1978).⁸ Our novel contribution to the existing literature lies in applying established clustering statistical techniques for functional data (Jacques and Preda, 2014) within the context of financial data analysis. We find that volatility (both BRV and BVIX) is an important variable for characterizing the drivers of the clusters. This finding aligns with several other studies that consider volatility as an important conditioning variable. For instance, Chabi-Yo (2012) demonstrate that PK increases with market volatility, while Song and Xiu (2016) and Linn, Shive, and Shumway (2018) show that the PK remains nonincreasing when consistently conditioned. Our work is also related to the conditional estimation of the pricing kernel and its link to the variance risk premium. In the equity index markets, Christoffersen, Heston, and Jacobs (2013) propose a U-shaped pricing kernel driven by a positive VRP, while Almeida and Freire (2022) use demand as a state variable and find that the pricing kernel is U-shaped and VRP is high when public investors are net-selling OTM options. Grith, Härdle, and Krätschmer (2017) report a hump-shaped pricing kernel under low VRP and a U-shaped pricing kernel under high VRP. Similarly, our findings in the Bitcoin market reveal a more pronounced hump in the LV regime, which is characterized by a lower BVRP.⁹

Forth, our paper is also related to the literature that addresses the drivers of cryptocurrency prices and risk premia. According to Biais et al. (2023), the fundamental value of Bitcoin is related to net transactional benefits on the platform. But its price fluctuates substantially due to uncertainty about the fundamentals, institutional risk, sentiment, speculation, or manipulation.

⁸In contrast, estimating the conditional physical density without making restrictive assumptions is widely recognized as challenging (Linn, Shive, and Shumway, 2018, Barone-Adesi et al., 2020).

⁹Apart from the variance or variance risk premium, other factors may play a significant role in explaining the variation in risk premia within the Bitcoin market. For instance, focusing on equity indexes, Rosenberg and Engle (2002) find a positive correlation between the PK's slope and recession indicators. At the same time, the slope negatively correlates with expansion signs for S&P 500 index, while Grith, Härdle, and Park (2013) observe that the PK hump is more pronounced during economic expansions than in recessions for the German DAX Index. A similar analysis focusing on fluctuations in the cryptocurrency market can also be performed for Bitcoin.

Cryptocurrencies – such as Bitcoin – facilitate peer-to-peer transactions on a digital platform supported by blockchain technology, which uses cryptography—hence the term "crypto." Biais et al. (2023) relate Bitcoin price to blockchain technology and its ecosystem. Further studies relate the value of cryptocurrencies to their adoption and evaluate the advantages of tokenization (Athey et al., 2016, Cong, Li, and Wang, 2021, Hinzen, John, and Saleh, 2022, Sockin and Xiong, 2023b, Sockin and Xiong, 2023a, Hautsch, Scheuch, and Voigt, 2015). For studies on price and volume manipulation, we refer to Makarov and Schoar (2020), Griffin and Shams (2020), and Cong et al. (2023); for sentiment and policy uncertainty, refer to Shen, Urquhart, and Wang (2019) and Demir et al. (2018). We investigate the predictability of BP and BVRP and their components, and find that the BP attributable to upside return states is predictable by sentiment—particularly that of bullish and individual investors—and investor attention.

2 Data

The data contains cash-settled European-style options traded on the Deribit exchange and daily Bitcoin prices, available via the [Blockchain Research Center](#) (BRC). Deribit is a margin-trading platform specializing in Bitcoin and Ethereum options, as well as futures and options. It has emerged as the leading global cryptocurrency options exchange. This platform was established in the Netherlands in June 2016 and was registered in Panama and then Dubai. In May 2025, Coinbase acquired it. Daily USD-denominated Bitcoin prices are collected after the early-adoption phase from January 2014 to December 2022 from Deribit.¹⁰ Deribit calculates Bitcoin settlement price as a weighted average across eleven major cryptocurrency exchanges over the last 30 minutes before settlement time (8 am UTC). Bitcoins are divisible, such that the quantity traded can be expressed in decimals, and are traded around the clock on several exchanges. The trading fees on Deribit are 0.03% of the underlying or 0.0003 Bitcoin per option contract, capped at 12.5% of the contract's value.

¹⁰We discard data prior to 2014 when Bitcoin prices were very volatile and less reliable. For instance, Bitcoin skyrocketed from \$13 at the beginning of 2013 to \$1000 by November of this year, increasing by over 75 times in just 11 months.

Table 1: Implied volatility of BTC options (Calls and Puts)

Moneyness	(0, 9]	[10, 26]	[27, 33]	>33	Average
< -0.6	4.36 (189)	2.20 (1,774)	1.84 (1,559)	1.29 (38,305)	1.36 (41,827)
[-0.6, -0.2)	1.56 (105,267)	1.19 (179,836)	1.05 (42,420)	0.95 (307,367)	1.12 (634,890)
[-0.2, 0.2]	0.81 (3,691,380)	0.77 (1,162,120)	0.76 (174,089)	0.79 (681,068)	0.80 (5,708,657)
(0.2, 0.6]	1.37 (99,445)	1.10 (203,733)	0.86 (52,479)	0.84 (313,562)	0.95 (669,219)
> 0.6	2.18 (5,899)	1.44 (28,666)	1.19 (14,869)	0.99 (304,534)	1.07 (353,968)
Average	0.85 (3,902,180)	0.86 (1,576,129)	0.86 (285,416)	0.88 (1,944,836)	0.86 (7,408,561)

Average implied volatility of Bitcoin options across moneyness and maturity (calls and puts combined). Columns correspond to time-to-maturity (in days). Implied volatilities are aggregated using contract-weighted averages. The number of observations (Bitcoin option contracts) is reported in parentheses.

We use daily transaction-level options data spanning from July 2017 to December 2022. The raw data includes the timestamp, order type (call or put), volume, instrument price, strike price, spot price (the price of the underlying), implied volatility, and transaction type (buy or sell). Each contract has a lot size of 1 Bitcoin. All prices and instruments are denominated in U.S. Dollars. We implement some filtering to mitigate potential errors in the raw data. A notable distinction in our paper is the nature of transaction data, which provides a single option price per transaction rather than separate bid and ask prices. Further, we exclude option observations where i) options whose implied volatility is zero or negative, ii) implied volatility is missing or non-positive, iii) no-arbitrage conditions are violated, iv) transaction quantity is non-positive. After filtering, our dataset comprises 1992 days, including 5,384,537 transactions. We report summary statistics of the implied volatility data across bins of moneyness¹¹ and maturity in Table 1, together with the average number of Bitcoin option contracts in each cell, which reflects the underlying data availability and liquidity across segments. We use the daily 1-month Treasury bill rate available on the FRED (Federal Reserve Bank of St. Louis) website as a proxy for the risk-free rate, and we set the cost-of-carry rate to zero.¹² Further data details are provided in Appendix A.1.

¹¹The moneyness of a contract is $m = K/S$, where K is the strike price, and S denotes the current Bitcoin price.

¹²We have also experimented with the zero interest rate – in accordance with the practices observed on the Deribit exchange – as a proxy for the risk-free rate. This choice makes virtually no difference to our calculations.

3 Theoretical Framework

Let the price process of the Bitcoin index be a stochastic process with continuously distributed marginals S_t under the physical measure $\mathbb{P}$, equipped with a filtration $\mathcal{F}_t$. In what follows, we focus on unconditional distribution of the τ -days ahead random returns $R = (S_{t+\tau} - S_t)/S_t$.¹³ Removing the time index indicates that we are not adopting a time-series approach.

The arbitrage-free assumption implies the existence of an equivalent measure $\mathbb{Q}$ (to $\mathbb{P}$) identified with a risk-neutral pricing rule. Under such a measure, discounted prices have the martingale property, such that the returns satisfy $\mathbb{E}_{\mathbb{Q}}(R) = R^f$, with R^f being the risk-free rate. Furthermore, we assume that the probability measures $\mathbb{P}$ and $\mathbb{Q}$ are differentiable with respect to the returns. Then, for each value r of the returns $p(r) = \frac{\partial \mathbb{P}(r)}{\partial r}$ and $q(r) = \frac{\partial \mathbb{Q}(r)}{\partial r}$, with $q(r)$ being the risk-neutral density and $p(r)$ being the physical density. In addition, we assume that $p(r) \neq 0$ for all possible returns $r \geq -1$. We utilize the information contained in the two pricing rules to analyze the first and second moment risk premia and gain insights into the pricing of risk.

Under the no-arbitrage condition, it holds that the unconditional return premia for Bitcoin (BP) is

$$\text{BP} := \mu_{\mathbb{P}} - \mu_{\mathbb{Q}}, \quad (1)$$

where $\mu_{\mathbb{P}} = \mathbb{E}_{\mathbb{P}}(R)$ and $\mu_{\mathbb{Q}} = \mathbb{E}_{\mathbb{Q}}(R)$. A positive BP premium is typically associated with investors seeking compensation for assuming directional risk. Equation (1) allows for a direct comparison with Beason and Schreindorfer (2022), with one significant difference: Bitcoin does not provide dividends.

To characterize risk pricing consistently with a more complex data-generating process that accommodates stochastic variance and jumps, we analyze the variance risk premium for Bitcoin (BVRP)

$$\text{BVRP} := \sigma_{\mathbb{Q}}^2 - \sigma_{\mathbb{P}}^2, \quad (2)$$

¹³Working with simple net returns enables a direct comparison to Beason and Schreindorfer (2022).

where $\sigma_{\mathbb{Q}}^2 = \text{Var}_{\mathbb{Q}}(R)$ and $\sigma_{\mathbb{P}}^2 = \text{Var}_{\mathbb{P}}(R)$. A positive BVRP indicates that variance buyers are willing to pay a premium to hedge away upward movements in the variance of the returns. In contrast, a negative BVRP indicates that the buyers would request a positive premium to long volatility. The BVRP in Equation (2) is defined differently than the BP in that a premium is paid to avoid the variance risk of the asset, hence the resulting sign inversion for the moments under the two measures.

3.1 Bitcoin Premium Decomposition

We utilize the method proposed by Beason and Schreindorfer (2022), originally used to analyze the S&P 500 market, to investigate the decomposition of BP in different return states, such that

$$\text{BP}(r) = \frac{\int_{-1}^r x \{p(x) - q(x)\} dx}{\text{BP}} \quad (3)$$

measures the fraction of the average Bitcoin first moment premium that is associated with returns below r . For ease of interpretation, we use a standardization by the BP that guarantees that the $\text{BP}(r)$ function approaches zero for returns in the far left tail and one for returns in the far right tail. Note that the $\text{BP}(r)$ function is not restricted to be monotonically increasing and can take values larger than one. Equation (3) indicates that $\text{BP}(r)$ increases when the risk-neutral density exceeds the physical density for negative return states, and when the physical density is greater than the risk-neutral density for positive return states. Economically, the increasing segments of $\text{BP}(r)$ are associated with states that contribute positively to the return premium.

The no-arbitrage assumption is also equivalent to the existence of a positive random variable π , called a stochastic discount factor (SDF), such that $\mathbb{E}_{\mathbb{P}}(R\pi) = R^f$. We refer to the projection of the SDF π on the set of Bitcoin returns as the pricing kernel function $\text{PK}(r) = \mathbb{E}[\pi|R=r]$ with $\mathbb{E}[\text{PK}(R)] = 1$. We define the pricing kernel associated with return r as the Radon-Nykodim derivative of the risk-neutral measure with respect to the physical measure

$$\text{PK}(r) = \frac{\partial \mathbb{Q}}{\partial \mathbb{P}}(r) = \frac{q(r)}{p(r)}. \quad (4)$$

$PK(r)$ reflects the risk price associated with the return r . Whenever the risk-neutral density exceeds the physical density, PK is larger than one, which means that the market risk price for that state is expensive relative to its (state) probability. Consequently, PK can provide crucial insights into the pricing rules of risk for the marginal investor across different return segments.¹⁴

In addition to the ratio-based estimator for the PK in Equation (4), we also use series-expansion estimators following Chabi-Yo and Loudis (2020), Chabi-Yo and Loudis (2024), and Schreindorfer and Sichert (2025). We refer the reader to these papers for implementation details.

3.2 Market Regimes and Risk-Neutral Density Clustering

In this article, we propose a nonparametric, data-driven approach for the conditional analysis of Bitcoin premia and the pricing kernel shape, highlighting their variations across different market regimes. We utilize risk-neutral densities to capture future market expectations and to identify similar market regimes. Our objective is to group these densities into homogeneous clusters. Due to the continuous nature of the data, we adopt a functional data approach and integrate information from both moneyness and time-to-maturity dimensions relevant to option prices.¹⁵

A straightforward method for clustering densities involves focusing on a specific time to maturity, which we refer to as the *univariate* clustering approach. While this method is simple to implement, it has one significant drawback: it overlooks the information contained in the term structure of options. This limitation can be addressed by using a more reliable method known as the *multivariate* clustering approach, which produces economically meaningful results. The *multivariate* approach considers a set of time-to-maturities (either continuous or discrete) that represent the term structure and treats the corresponding risk-neutral densities as a series of curves. This method is preferred over the *univariate* approach because it incorporates information across both the moneyness and expiry dimensions, leading to more reliable clustering outcomes.

¹⁴Most studies define the SDF in general equilibrium models as a function of aggregate consumption or wealth, often assuming a representative agent with expected utility preferences. A common assumption is that SDF is proportional to the marginal utility, decreasing as consumption rises. Consequently, under certain conditions, such as co-monotonicity between consumption and returns, the pricing kernel $PK(r)$ is expected to decrease with returns.

¹⁵Functional data consists of continuously observed curves. For a general overview of functional data, we refer to Wang, Chiou, and Müller (2016). For more information on multivariate functional data methods, see Koner and Staicu (2023).

Additionally, it allows for a more nuanced representation of investors' future expectations.

Although the implied volatility surface has the potential to be used for clustering, our empirical investigation indicates that employing risk-neutral densities (RNDs) leads to more stable clustering outcomes.¹⁶ This stability can be attributed to the fact that RNDs are closely connected to the second derivative of the call function for a fixed maturity (as discussed in Section 4.1.1). Prior research in functional data analysis shows that second derivatives are more responsive to changes in the latent processes that drive variability in the observed data (Grith et al., 2018). In contrast, implied volatilities involve a simpler nonlinear transformation from option prices to volatilities.

To effectively differentiate between these functional objects, one needs to compute a distance metric and then apply a clustering algorithm. In the context of clustering and classifying functional data, hierarchical clustering and the k -means partitioning methods are two classical and widely used techniques; see Jacques and Preda (2014) for a review on clustering. We use the default L^2 distance in the multivariate function space to apply hierarchical clustering, where the mean functions are defined as the centers of each cluster.

However, because density functions satisfy the constraints $\int f(x)dx = 1$ and $f \geq 0$, they are not situated within a vector space. Consequently, traditional functional data analysis methods based on Hilbert space are not applicable (Petersen and Müller, 2016). An isomorphic mapping of the densities to the standard L^2 space is required to perform standard statistical Hilbert space methods. Different transformations are possible, such as taking the natural logarithm. As outlined in Machalova, Hron, and Monti (2016) and Eckardt, Mateu, and Greven (2022), a straightforward isomorphism that has shown better results in practice is the centered-log-ratio (CLR) transformation. The transformation is applied to the RND function and is defined as

$$\text{clr}\{q(r)\} = \log \left\{ \frac{q(r)}{\mu_G} \right\}, \quad (5)$$

with the geometric mean of the risk-neutral density function given by $\mu_G = \exp[\mathbb{E}\{\log(q(r))\}]$.

¹⁶Clustering results based on implied volatilities are available upon request.

In the second step, we compute the L^2 distance between all pairs of transformed density sets indexed by i and j , for a continuum of τ . The distance between two-dimensional functions is defined as:

$$D(i, j) = \sqrt{\int_{\tau} \int_r [\text{clr}\{q_i(r, \tau)\} - \text{clr}\{q_j(r, \tau)\}]^2 dr d\tau} \quad \text{for all } i, j,$$

where $i = 1, \dots, T$ and $j = 1, \dots, T$. Building upon the resulting Euclidean distance matrix, the risk-neutral densities are grouped into homogeneous clusters, where homogeneity is measured by the symmetric distance measure of the transformed densities.

We applied hierarchical clustering using Ward's criterion (Ward Jr, 1963), which merges clusters by minimizing the increase in within-cluster variance. This approach was chosen because it typically yields compact clusters, in which observations within the cluster are very close in a geometric sense. In addition, Ward's method yields a more stable hierarchical structure than linkage criteria derived from pairwise distances, which are sensitive to outliers.

4 Estimation Procedure

To integrate existing approaches that combine option prices and underlying asset returns to estimate and decompose risk premia, we employ flexible estimation procedures. All estimators are semi- or nonparametric, allowing us to capture the underlying structure of the data without imposing restrictive functional forms.

4.1 Densities Estimation

4.1.1 Risk Neutral Density

Interpolation of Implied Volatility Surface. The estimation of the risk-neutral density (RND) involves several carefully designed steps, outlined below. First, we need to estimate a smooth implied volatility surface. To accomplish this, we employ a smoothing technique tailored to the data structure, which leverages semi-parametric models. These models impose no-arbitrage restric-

tions and provide reasonable extrapolation to the tails, where observations are sparse. Second, the smooth surfaces are adopted to estimate a daily set of RNDs for the desired option expiry dates. The procedure for obtaining the empirical smooth implied volatility, $\hat{\sigma}(K, \tau) = \sqrt{\omega_{\hat{\theta}_t|\tau}(K/S_t)}$, is described in Section A.3 of the Appendix.

After generating the SVI-smoothed IV curves for each available maturity on a given day, we interpolate linearly along the maturity dimension—using a weighted average of the two closest maturities to construct IV curves of unobserved maturities.¹⁷ After the smoothing step, we fix a discrete set of maturities, focusing on the short leg of the options term structure, with 9, 27, and 45 days to maturity, for which we estimate the RNDs.

Estimation of the Risk-Neutral Density. In the next step, we rely on the classic theoretical result by Breeden and Litzenberger (1978) to recover the RND for a given maturity of options. Through a change of variable, the RND is represented as a function of returns

$$q_t(r) = S_t e^{R_t^f \tau} \frac{\partial^2 C_t}{\partial K^2}. \quad (6)$$

Nonparametric estimation of the risk-neutral density (RND) requires a smooth call function based on the strike price. In our implementation, we utilize the smooth implied volatility surface obtained in the previous step to convert the implied volatilities into call prices using the Black-Scholes pricing formula: $\hat{C}_t(K, \tau) = C_{BS}(K, \tau, S_t, R_t^f, \hat{\sigma}_\theta(K, \tau))$. We then compute the second derivative numerically to obtain the conditional RND estimates. Since the resulting densities may not integrate to one, we normalize each estimate by its integral. This ensures that each estimate is a valid probability density function $\hat{q}_t(r)$. Finally, for a fixed τ , we collect a series of daily estimated RNDs indexed by t and define the unconditional RND as a sample average over the entire sample

$$\hat{q}(r) = \frac{1}{T} \sum_{t=1}^T \hat{q}_t(r). \quad (7)$$

¹⁷This approach to smoothing the surfaces is more flexible than the one used by Beason and Schreindorfer (2022), who assume linearity in the SVI parameters with respect to τ and fit the model by pooling the observations along the moneyness and time-to-maturity dimensions. Further details about the interpolation are provided in Appendix A.3, and the interpolated IV surface is shown in Figure A3.

4.1.2 Physical Density

To estimate the physical density, we follow Beason and Schreindorfer (2022). We first construct the empirical PDF as a histogram of the full sample of overlapping returns. The central region, between the 10th and 90th percentiles, is then smoothed using a 10th-order polynomial, while the tails are modeled using a Generalized Extreme Value (GEV) distribution. Details on the estimation of the unconditional density, $\hat{p}(r)$, are provided in Section A.4 of the Appendix.

4.2 Risk Premia Estimation

Empirical Bitcoin Premium: $\widehat{\mathbf{BP}} = \widehat{\mu}_{\mathbb{P}} - \widehat{\mu}_{\mathbb{Q}}$. It is the unconditional premium involved in estimating the first moment of the returns. The point estimate for the first moment premium is known to be a noisy measure. Therefore, we have experimented with several estimators. For the empirical annualized first moment under the $\mathbb{P}$ measure, we use the following estimator $\widehat{\mu}_{\mathbb{P}} = \frac{365}{27} \int_{-1}^1 x \{\hat{p}(x)\} dx$.¹⁸ Results for the sample mean are available in Table B5. We also computed the option-based first moment lower bounds following Martin (2017) and Chabi-Yo and Loudis (2020). As discussed in the data section, we use the 1-month (annualized) Treasury bill rate as a proxy for the risk-free rate.

Empirical Variance Risk Premium: $\widehat{\mathbf{BVRP}} = \widehat{\sigma}_{\mathbb{Q}}^2 - \widehat{\sigma}_{\mathbb{P}}^2$. Estimating the variance risk premium is a primary focus of this work. We rely on the Bollerslev-type variance risk premium framework (Bollerslev, Tauchen, and Zhou, 2009, Bollerslev, Gibson, and Zhou, 2011, Bollerslev and Todorov, 2011), in which the conditional variance risk premium is defined as the difference between the risk-neutral expected variance and the expected variance under the physical measure,

$$\mathbf{VRP}_t = \mathbb{E}_t^{\mathbb{Q}}(\sigma_{t+1}^2) - \mathbb{E}_t^{\mathbb{P}}(\sigma_{t+1}^2).$$

Building on earlier studies, we employ options-based volatility as a proxy for the forward $\mathbb{Q}$ -volatility. We calculate the daily volatility Index, the BVIX, for different tenors. Our method-

¹⁸An alternative estimate of the unconditional mean under the $\mathbb{P}$ measure is given by the sample mean, $\hat{\mu} = \frac{1}{T} \sum_{t=1}^T R_t$. The resulting estimates are consistent across the two approaches; see Tables B5 in the Online Appendix.

ology for constructing the daily $BVIX_t$ utilizes intraday option data on Bitcoin. It builds on the fair pricing of variance swaps employed by the CBOE to compute the Volatility Index (VIX). The squared BVIX reflects a market-specific expected Bitcoin variance directly captured from options, and we use its sample mean $BVIX = \frac{1}{T} \sum_{t=1}^T BVIX_t$ to compute $\hat{\sigma}_{\mathbb{Q}}^2 = BVIX^2$ as a proxy for the $\mathbb{Q}$ -variance. Details on the calculation of the BVIX are provided in Appendix A.2.

To estimate the unconditional physical centered second moment, we first obtained a series of (annualized) monthly realized variances $RV_t^2 = \frac{365}{27} \sum_{l=1}^{\tau} r_{d,t-l}^2$, $r_{d,t} = \log S_t / S_{t-1}$, calculated as the sum of squared log returns over the past 27 days. We estimate the conditional volatility of log returns based on the heterogeneous autoregressive (HAR) model of Corsi (2009):

$$RV_{t+h} = \alpha_h + \beta_h^m RV_t^{(27)} + \beta_h^w RV_t^{(5)} + \beta_h^d RV_t^{(1)} + \varepsilon_{t+h},$$

where $RV_t^{(\tau)} = \left(\frac{1}{\tau} \sum_{j=0}^{\tau} \log(1 + R_{t-j})^2 \right)^{0.5}$. Weekly ($RV_t^{(5)}$) and monthly ($RV_t^{(21)}$) volatility measures allow the model to capture the long-memory feature of stock market volatility. We then take the mean $\hat{\sigma}_{\mathbb{P}}^2 = \frac{1}{T} \sum_{t=1}^T \widehat{RV}_{t+h}^2$ over the entire sample.¹⁹

We note that this approach does not allow us to disentangle the diffusive and jump components of variance. Existing studies show that, under certain conditions, realized variance converges to the quadratic variation of returns, which includes both integrated variance and jump variation.²⁰ Consequently, Bollerslev-type estimators do not separately identify continuous volatility risk and jump risk; instead, both components are embedded in realized variance and enter the estimated variance risk premium jointly.²¹

¹⁹As an alternative approach, variance is estimated, by integrating the squared deviation of returns from the mean over the respective risk-neutral density or physical density for a specified maturity $\hat{\sigma}_{\mathbb{Q}}^2 = \frac{365}{\tau} \int_{-1}^{\infty} \left\{ x - \int_{-1}^{\infty} z \hat{q}(z) dz \right\}^2 \hat{q}(x) dx$ and $\hat{\sigma}_{\mathbb{P}}^2 = \frac{365}{\tau} \int_{-1}^{\infty} \left\{ x - \int_{-1}^{\infty} z \hat{p}(z) dz \right\}^2 \hat{p}(x) dx$. All volatility measures are annualized for consistency. These results are available upon request.

²⁰To note that these studies typically compute realized variance using intraday data by summing squared high-frequency returns within each trading day. In contrast, due to data limitations, we rely on daily returns, implying that our realized variance measure should be interpreted as a lower-frequency and therefore noisier approximation of the integrated variance.

²¹Consistent with this limitation, our analysis does not separately identify diffusive variance risk and jump risk premia. A natural extension would be to disentangle these components and isolate the contribution of jump risk to the overall variance risk premium. Implementing such a decomposition in the Bitcoin market is beyond the scope of this paper, as it requires high-frequency data on the underlying index in addition to option prices, as well as more intricate econometric procedures to identify and measure jumps (Bollerslev, Todorov, and Xu, 2015; Kolokolov, 2023; Todorov, 2010). We leave this analysis to future research.

Bitcoin Premium Decomposition and Pricing Kernel. The Bitcoin premium decomposition is given by Equation (3). We implement it for the monthly returns. The pricing kernel is estimated following Equation (4) using the empirical risk-neutral density $\hat{q}(r)$ as defined in Section 4.1.1 and the physical density $\hat{p}(r)$ defined in Section 4.1.2 on the full sample of overlapping monthly returns.

4.3 Conditional Estimation of Risk Measures

The unconditional estimators presented in Section 4.2 rely on averaging time series quantities. Similarly, for the conditional estimators, we average observations belonging to each cluster. While the conditional estimates at any particular point are expected to be noisy, the unconditional and conditional ones based on clustering are expected to be more stable. Relying on two clusters that describe highly volatile (HV) and less volatile (LV) market regimes, as shown in Section 4.4, we define conditional estimators below.²²

The conditional (on clusters) variances are $BVIX_i^2 = \frac{1}{|C_i|} \sum_{t \in C_i} BVIX_t^2$ and $RV_i^2 = \frac{1}{|C_i|} \sum_{t \in C_i} RV_t^2$, where each cluster i is represented as a set of dates $C_i = \{t | t \in \text{Cluster } i\}$ for $i = HV, LV$. Similarly, the risk-neutral density for the clusters is given as $\hat{q}_i(r) = \frac{1}{|C_i|} \sum_{t \in C_i} \hat{q}_t(r)$, for $i = HV, LV$.

To estimate the conditional physical density across volatility regimes, we adopt a shape-invariant approach based on rescaled returns (Lawton, Sylvestre, and Maggio, 1972, Kneip and Engel, 1995). The key idea is that return distributions differ across regimes primarily through their scale (volatility), while their standardized shape remains approximately stable. Formally, for each observation t belonging to regime $i \in \{HV, LV\}$, we construct rescaled returns as

$$r_{t,i} = \frac{RV_i}{RV} r_t,$$

where RV_i denotes the average realized variance in regime i , and RV is the unconditional realized variance. This transformation rescales returns using regime-specific volatility, thereby allowing us to use the full sample while accounting for differences in volatility across regimes.

²²The hierarchical clustering algorithm used indicates two main clusters. We display the two main modes of variation and cluster allocation of the densities in a low-dimensional representation in Section A.6 in the Appendix.

Under this shape-invariance assumption, the conditional physical density can be estimated by applying the same nonparametric procedure described in Section 4.1.2 to the rescaled returns $r_{t,i}$. The resulting densities $\hat{p}_i(r)$ capture regime-specific dispersion while preserving the underlying shape of the return distribution. These densities are then used to compute conditional moments, risk premia, and their decomposition, and the conditional pricing kernels.

4.4 Clustering of Risk Neutral Densities

The proposed classification is based on the endogenous variation of risk-neutral measures. We first estimate the risk-neutral density as described in Section 4.1.1 and apply the CLR transformation of Equation (5). For the multivariate clustering approach introduced in Section 3.2, we focus on maturities $\tau = 9, 27$, and 45.

The choice of the number of clusters is guided by the low-dimensional structure of the data. Specifically, we examine the first two principal components of the distance matrix, shown in Figure A5a, which reveal a clear separation into two groups. We further validate this result using Uniform Manifold Approximation and Projection (UMAP) (McInnes et al., 2018), which captures non-linear dependencies between risk-neutral densities (see Appendix A.6). The UMAP embedding, reported in Figure A5b, exhibits a consistent two-group structure.

Taken together, these results provide strong empirical support for selecting two clusters, corresponding to distinct regimes in the cross-section of risk-neutral densities.

5 Empirical Results

Our research focuses on a 27-day investment horizon. Table 2 summarizes the main results for the unconditional and conditional BP and BVRP estimates. The BP is significantly higher than that of traditional investment assets such as currencies, commodities, and stocks, averaging around 66% per year. It is known that the point estimate for the first moment risk premium is a noisy measure. For this reason, we also estimate first-moment premium lower bounds for the BP based on measures extracted from the options data. The lower bounds reflect the minimum compensa-

tion investors require for holding Bitcoin. The BP lower bound of Martin (2017) requires only knowledge of the risk-neutral variance, while the lower bound of Chabi-Yo and Loudis (2020) employs skewness and kurtosis as well. We report the results in Table A6 in Appendix A.7. We find that the average lower bound for annualized monthly BP is 66% for the former and the unrestricted estimate for the latter is 82%. The annualized risk-neutral and physical monthly variances, proxied by the squared Bitcoin Volatility Index (BVIX) and the realized volatility (BRV), are significantly high: 0.72 and 0.50, respectively. The corresponding variance risk premium is 0.22, much higher than that of the S&P 500 index—approximately 2%, according to Bollerslev, Tauchen, and Zhou (2009).

Table 2: Risk Premia

Panel A: Bitcoin Premium			
	Overall	HV	LV
$\widehat{BP} = \widehat{\mu}_P - \widehat{\mu}_Q$	0.66	0.69	0.62
$\widehat{\mu}_P$	0.67	0.70	0.63
$\widehat{\mu}_Q$	0.01	0.01	0.01
Panel B: Bitcoin Variance Risk Premium			
	Overall	HV	LV
$\widehat{BVRP} = \widehat{\sigma}_Q^2 - \widehat{\sigma}_P^2$	0.22	0.30	0.07
$\widehat{\sigma}_Q^2$	0.72	0.86***	0.46***
$\widehat{\sigma}_P^2$	0.50	0.56***	0.39***
Days	1017	649	368

Panel A: Estimates of the unconditional BP and conditional BP_i . **Panel B:** Estimates of the unconditional BVRP and conditional BVRP_{*i*}. Unconditional estimates are referred to as 'Overall', and the conditional ones are cluster-specific for $i \in \{HV, LV\}$. $\widehat{\mu}_P = \frac{365}{27} \int_{-1}^1 x \{ \hat{p}(x) \} dx$ and $\widehat{\sigma}_P^2(R)$ is the sample mean of the forecasted (HAR) variance. $\widehat{\mu}_Q$ is estimated using 1-month Treasury bill rates and for $\widehat{\sigma}_Q^2(R)$ we use square of BVIX. All the estimates are annualized. ANOVA is applied to test whether the conditional estimates are different than the unconditional ones (H_0 : no difference), with 1%(***) , 5%(**) and 10%(*) denoting significance level.

We further analyze conditional estimates across market regimes. Our results indicate that BP is higher in the HV regime. Martin (2017) lower bounds are estimated at 78% in the HV regime and 44% in the LV regime, while Chabi-Yo and Loudis (2020) bounds are 95% in the HV regime and 56% in the LV regime. These results indicate that in volatile markets, the BP may be higher than our point estimate in Table 2. Furthermore, risk-neutral and physical variances exhibit substantial variation across clusters. Specifically, the HV cluster describes a highly volatile market, characterized by high second moments of Bitcoin returns, with monthly annu-

alized variances of 0.86 for the risk-neutral and 0.56 for the physical measure. In contrast, the LV cluster describes a less volatile market, identifiable by smaller variance proxies, with a risk-neutral variance of 0.46 and physical variance of 0.39. The overall increase in volatility leads to higher option price premia on average across all levels of moneyness.²³ The variances under the two probability measures differ substantially, introducing a substantial variance risk premium in both market regimes. The low-volatility cluster is characterized by a much lower BVRP of 0.07, compared to the high-volatility cluster's 0.30. By relying on the average values of the premia within each cluster, we observe a positive relationship between the BP and BVRP in the Bitcoin market. Simultaneously, our clustering results indicate a positive relationship between returns and variance, supporting the inverse leverage effect in the Bitcoin market documented by Hou et al. (2020).²⁴

The unconditional decomposition of the $BP(r)$ exhibited in Figure 1 reveals that $BP(r)$ is mostly increasing for return states smaller than -20% and larger than 20%, reflecting that these states contribute positively to the Bitcoin premium. BP attributable to positive and negative returns has a similar share, with positive returns contributing slightly more to the BP. For illustration, we focus on return states that roughly span from one to three standard deviations away from the mean.²⁵ We find that negative monthly returns between -60% and -20% and positive returns ranging from 20% to 60% contribute 28.53% and 38.91%, respectively of the BP. The comparable attribution of the Bitcoin premium between positive and negative returns contrasts with the results in the S&P 500 index market, where 80% of EP is attributable to monthly returns below -10%, as reported by Beason and Schreindorfer (2022), while positive returns have a much smaller (positive) share.²⁶

²³Higher option prices in the HV regime are also shown as a higher implied volatility surface in Figure A3 in the Appendix.

²⁴Hou et al. (2020) find a positive correlation coefficient between the increments in the return and volatility equations of the stochastic volatility with a correlated jump (SVCJ) model of Duffie, Pan, and Singleton (2000). This implies that increasing prices are associated with increasing volatility. We find that the first moment in the Bitcoin market rises during periods of high realized variance. This suggests that BP increases following periods of increased volatility.

²⁵For the Bitcoin index, the standard deviation is 0.22 for the unconditional physical distribution of monthly returns. By contrast, the results in Beason and Schreindorfer (2022) suggest that the standard deviation for S&P 500 monthly returns is 0.04.

²⁶Beason and Schreindorfer (2022) primarily focus on negative returns and do not quantify the risk premia associated with positive returns. From their Figure 1, we can determine that the positive equity premium is linked to returns

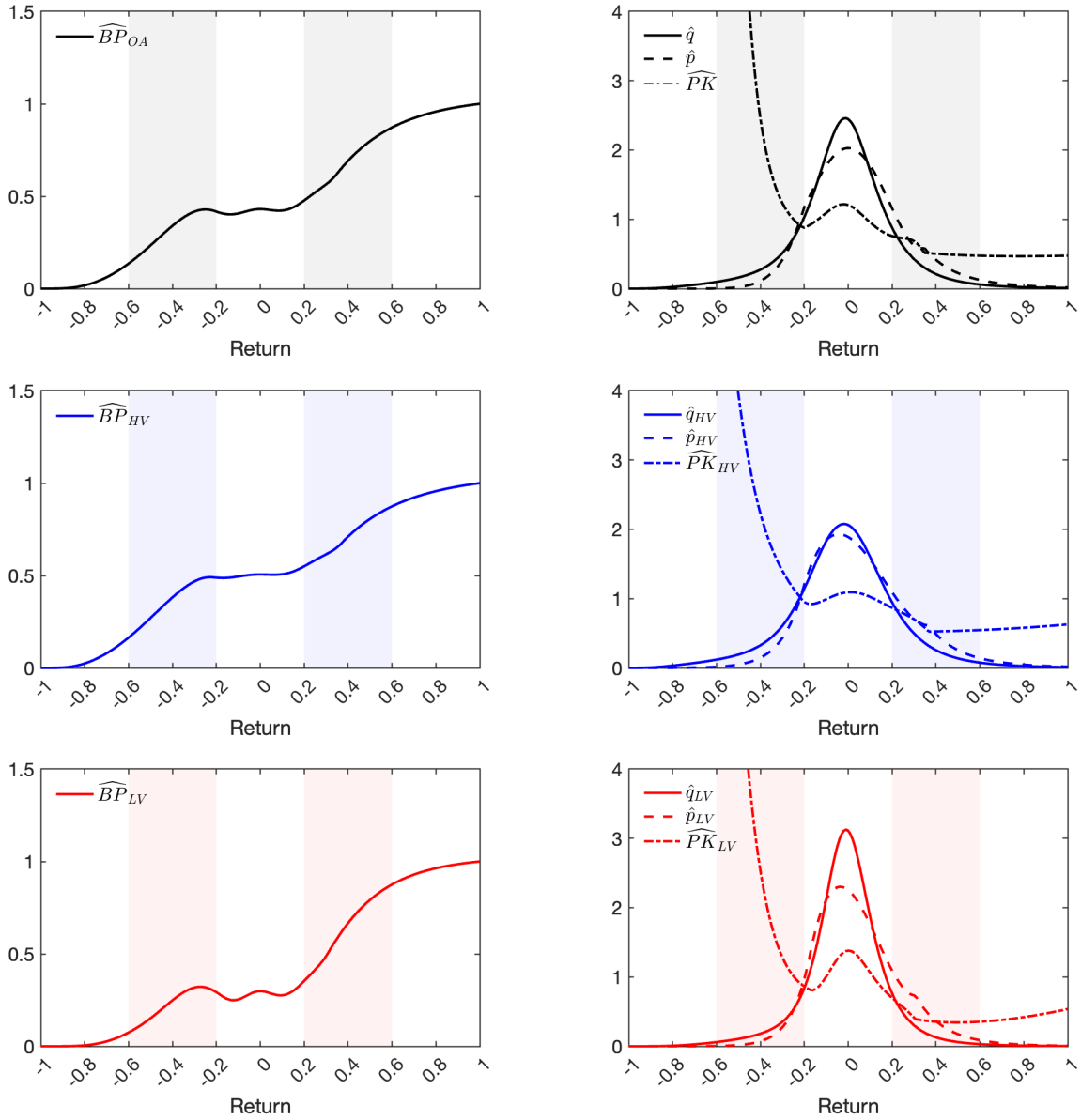


Figure 1: First column includes estimated BP for overall (black), HV cluster (blue), and the LV cluster (red). The second column includes estimated PK, physical, and risk-neutral density for overall (black), HV cluster (blue) and the LV cluster (red). The shaded areas mark the returns range $[-0.6, -0.2]$ and $[0.2, 0.6]$. **First row.** For the overall sample, the shaded areas contribute 28.53% and 38.91% to the overall BP, respectively. **Second row.** For the HV regime, the shaded area contributes 32.54% and 31.86% to the BP. **Third row.** The shaded areas of the LV cluster contribute 22.52% and 51.55% to the BP.

The $PK(r)$ is generally decreasing, except for a segment of Bitcoin returns between -20% and 0%, which creates a hump. The hump is above one, indicating that ATM options are expensive, between 0% and 5%, and accounts for approximately 25% of the overall equity premium. In addition, for S&P 500 returns between 5% and 20%, the equity premium is decreasing, and these return states contribute negatively to the first moment premium.

leading to negative returns for both ATM put and call options. This also highlights the presence of a variance risk premium.²⁷

Next, we examine closely the risk functions across different market conditions. Figure 1 also displays the $BP(r)$ for the two clusters, revealing several noteworthy characteristics. Notably, the shape of $BP(r)$ in the HV cluster is more akin to the unconditional $BP(r)$. But there are also some important differences. The BP attributable to negative and positive returns is more balanced, with a share of 32.54% for returns between -60% and -20% and returns from 20% to 60% contributing only 31.86% to the BP. In the LV cluster, $BP(r)$ exhibits a substantial increase in the region of positive returns. We identify a novel pattern for returns ranging from 20% to 60% that exhibit a significant 51.55% positive contribution to the BP. In contrast, the share of BP attributable to negative returns ranging from -60% to -20% is only 22.52%. The BP attributable to returns below -60% and above 60% is relatively small. These results suggest that large positive returns, but not extreme returns, are the main source of Bitcoin risk premia during calm markets. Investors are more concerned with the risk compensation for the upside risk and less concerned with hedging the downside risk.

Table 3: Characteristics of BP, $\mathbb{Q}$ and $\mathbb{P}$ in influential states

	Negative states			Positive states		
	BP(-0.2)-BP(-0.6)	$\int_{-0.6}^{-0.2} p(r)dr$	$\frac{\int_{-0.6}^{-0.2} q(r)dr}{\int_{-0.6}^{-0.2} p(r)dr}$	BP(0.2)-BP(0.6)	$\int_{0.2}^{0.6} p(r)dr$	$\frac{\int_{0.2}^{0.6} q(r)dr}{\int_{0.2}^{0.6} p(r)dr}$
Overall	28.53%	0.09	1.44	38.91%	0.19	0.62
HV	32.54%	0.09	1.68	31.86%	0.08	0.66
LV	22.52%	0.07	1.42	51.55%	0.18	0.47

BP(-0.2)-BP(-0.6) and BP(0.2)-BP(0.6) report the percentage contribution of each return region to the Bitcoin premium. $\int p(r)dr$ denotes the physical probability over these states and $\frac{\int q(r)dr}{\int p(r)dr}$ is the associated risk price.

The results in Table 3 show the probabilities of large price movements away from the current value. The probability of an increase in returns between one and three standard deviations, is on average, twice as large as the probability of a decrease. Risk compensation on the left and right segments is balanced for the unconditional and the turbulent markets. In contrast, during less

²⁷Expensive ATM options have been used in the literature to document the variance risk premium (Coval and Shumway, 2001).

volatile markets, the main source of risk premia for Bitcoin is large positive returns. Their probability is 18%, double that of the high-volatility regime. These results show that large positive returns occur during less volatile markets.

We also calculate the price of risk as the ratio of the average risk-neutral density to the physical density, as in Beason and Schreindorfer (2022), which allows us to draw a comparison to the S&P 500 market. For negative states between -60% and -20%, the risk price for Bitcoin is approximately 1.44, which is lower than the 2.63 for the S&P 500 as reported by Beason and Schreindorfer (2022) for returns -30% and -10%, but comparable to levels found in Campbell and Cochrane (1999), Bansal and Yaron (2004), Barro (2009), and Wachter (2013).²⁸ These results show that the Arrow-Debreu prices are smaller on average in the Bitcoin market than in the S&P 500 market, as investors are willing to pay less for hedging the downside risk. Our analysis extends to the positive returns and finds a price of risk of 0.62 for returns between 20% and 60%; a direct comparison with the above-mentioned studies is not available.

We extend the analysis in Table 4, where we report the Bitcoin return premium and variance risk premium based on the decomposition of Chabi-Yo and Loudis (2024), for the entire period and for the high- and low-volatility regimes. Figure 2 displays the time series of the total conditional return premium together with its decomposition into downside, center, and upside components. In addition, in Figure A6 in the Appendix, we display the BP and BVRP estimates over time that rely on Chabi-Yo and Loudis (2020), Martin (2017), and Schreindorfer and Sichert (2025).

The average total return premium estimate is 0.69, broadly consistent with our nonparametric estimates. As in Table 2, the return premium is higher in the high-variance regime across all return regions. Downside and upside return regions account on average for 62% and 38% of the total premium, respectively, while the center region contribution is on average zero. In relative terms, the contributions of the three regions to the risk premia remain largely stable over different regimes.

²⁸A direct comparison on the same segment -30% and -10% would produce even lower prices for risk for the Bitcoin market on this segment. Similarly, calculating the price of risk for S&P 500 for segments beyond -30% would result in higher risk prices for large returns.

Table 4: Risk Premia Decomposition across Return Regions

Panel A: Bitcoin Premium (BP)						
	Overall		HV		LV	
	Level	%	Level	%	Level	%
$\widehat{BP}$	0.69	100	0.79	100	0.59	100
down	0.43	62	0.48	61	0.37	63
center	0.00	0	0.00	0	0.00	0
up	0.26	38	0.31	39	0.21	36
Panel B: Variance Risk Premium (VRP)						
	Overall		HV		LV	
	Level	%	Level	%	Level	%
$\widehat{VRP}$	0.06	100	0.05	100	0.06	100
down	-0.16	-267	-0.24	-480	-0.08	-133
center	-0.04	-67	-0.06	-120	-0.02	-33
up	0.26	433	0.36	720	0.16	267

Decomposition of Bitcoin risk premia across return states following Chabi-Yo and Loudis (2024). **Panel A** reports the Bitcoin premium (BP) and **Panel B** the variance risk premium (VRP). Return regions are defined as down $(-\infty, -0.1]$, center $(-0.1, 0.1]$, and up $(0.1, \infty)$. Results are reported in levels and as percentages of the total premium.

The average total BVRP is 0.06, much smaller than the point estimate in Table 2, but still way above the S&P 500 VRP. Over time, the BVRP stays positive and relatively constant across regimes. However, there are significant differences across return states. We find that the downside and central regions contribute negatively to the BVRP, while the upside region contributes positively. Moreover, the magnitude of the upside component exceeds that of the downside component. In relative terms, in the low-volatility regime, the contribution of the upside region (0.16) is approximately twice as large in magnitude as that of the downside region (-0.08).

Our findings contrast with Feunou, Jahan-Parvar, and Okou (2018), who document for the S&P 500 that the downside VRP is typically positive, reflecting compensation for downside risk, whereas the upside VRP is negative, consistent with a discount for upside exposure. Our findings also contrast with Chabi-Yo and Loudis (2024), who report a negative VRP in the downside region (which corresponds to a positive VRP in our framework) for the S&P 500. They further show that the central and upside regions contribute little to second-order risk premia, with estimates close to zero. In contrast, we find economically significant contributions from both tail regions. Finally, our findings are consistent with Lucic and Sepp (2024), who document that call

options in cryptocurrency markets tend to be relatively expensive, reflecting strong demand for upside exposure and implying a positive contribution of upside states to the VRP.

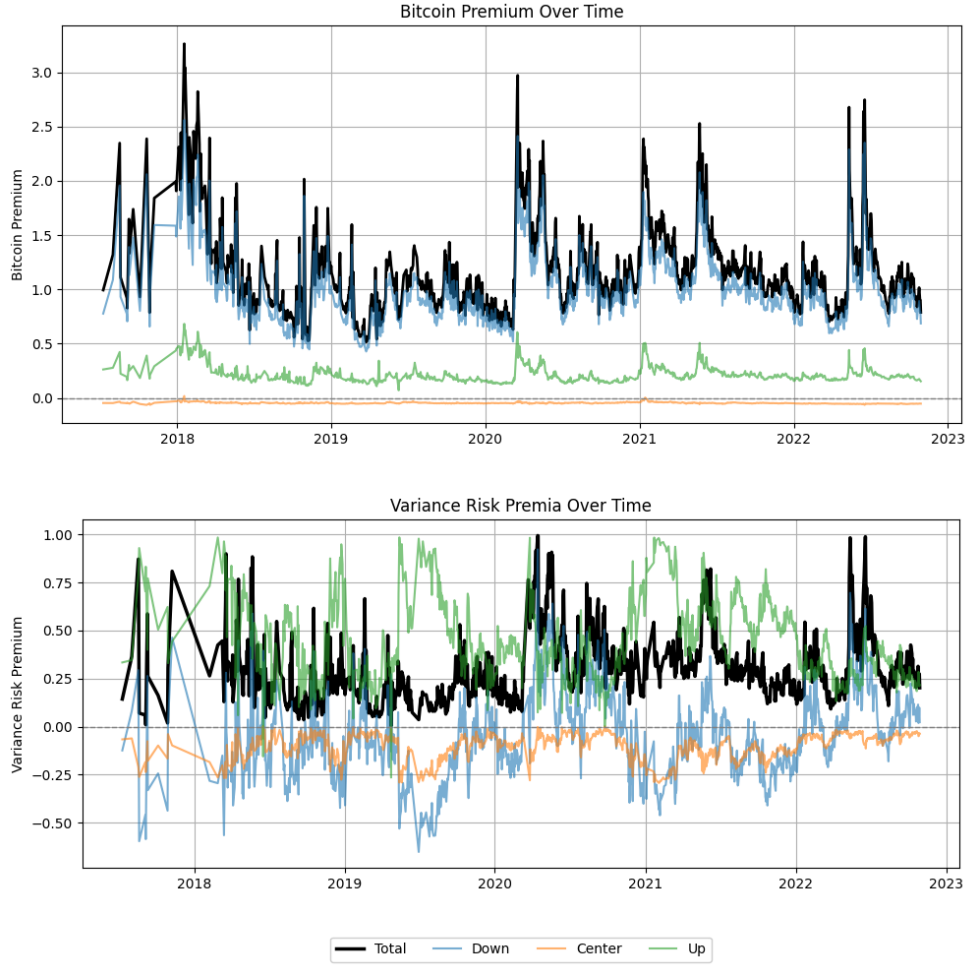


Figure 2: Time-varying decompositions of Bitcoin risk premia. Risk premia estimates rely on the conditional preference-based decomposition in Chabi-Yo and Loudis (2024). The figures show the total annualized BP and BVRP over time as well as the contribution of each return region, defined as down $(-\infty, -0.1]$, center $(-0.1, 0.1]$, and up $(0.1, \infty)$.

To investigate the robustness of our findings, we perform two further investigations. First, we extend the risk premia analysis to include 9 and 45-day horizons and investigate the robustness of our findings in Section A.8. The term structure of Bitcoin risk premia exhibits a downward-sloping pattern for the BP and an upward-sloping pattern for the BVRP across both volatility regimes, with these patterns more pronounced in the high-volatility regime, where both premia are consistently higher at all maturities. Second, we also apply our methodology to Ethereum data and uncover patterns similar to those observed for Bitcoin, albeit with quantitatively different

magnitudes. These results reported in Section A.10 suggest that the features we report also hold for other cryptocurrency markets.

5.1 Drivers of Bitcoin Risk Premia

We construct a comprehensive set of predictors for Bitcoin risk premia and their components, following Chabi-Yo and Loudis (2024). These variables span four broad categories: (i) macroeconomic conditions, general uncertainty, and crypto-market volatility; (ii) Bitcoin investor sentiment; (iii) momentum and investor attention; and (iv) trading activity and option-implied trading signals. We describe the data in more detail in Section A.9 of the Appendix.

Table 5: Predictability of Bitcoin premia across return regions

Dependent Variable	Predictor	Coef.	SE	R^2	N
Panel A: Bitcoin Premium (BP)					
Total	CRIX	−0.84***	0.34	0.00	894
	Google Trends (Bitcoin)	0.72*	0.41	0.05	59
Up	ADS Index	0.55***	0.20	0.01	193
	CRIX	−0.06**	0.02	0.01	1335
	Google Trends (Bitcoin)	0.15**	0.07	0.08	894
	Sentiment: Individual Investors	2.96**	1.36	0.01	59
	Sentiment: Bullish	5.59**	2.54	0.01	59
	Sentiment: Headline Index	2.90**	1.34	0.01	59
Down	CRIX	−0.23*	0.12	0.00	894
Panel B: Variance Risk Premium (VRP)					
<i>No statistically significant predictors</i>					

Each row reports the slope coefficient from a separate univariate predictive regression of one period ahead changes in the Bitcoin premia and their components on the previous period changes in the indicated predictor. Coefficients and standard errors for ADS Index are multiplied by 100, and for CRIX are multiplied by 10000. Standard errors are reported in a separate column. Sample sizes vary across regressions due to differences in data availability and frequency of the predictors. All values are rounded to two decimals. Significance levels: *** 1%, ** 5%, * 10%.

In Table 5, we report only statistically significant regression results. We find evidence of meaningful predictability of the Bitcoin premium (BP), with a pronounced asymmetry across return regions. Predictability is almost entirely concentrated in the upside component, whereas the downside component exhibits only weak and sporadic significance, and no statistically significant predictors are identified for the variance risk premium (VRP).

Across BP return states, CRIX emerges as the only predictor with consistent explanatory power, exhibiting a negative association with BP across total, upside, and downside components. This suggests that higher cryptocurrency market volatility is associated with lower expected return premia over all returns segments. Measures of investor attention and sentiment are significant only for the upside component. The economic magnitudes are substantial, indicating that periods of elevated attention and optimistic sentiment are associated with higher expected returns in positive return states. This pattern highlights the importance of speculative demand and attention-driven trading in cryptocurrency markets. In particular, measures of individual investor sentiment and bullish expectations exhibit significant positive predictive power, suggesting that optimistic sentiment is associated with higher subsequent upside risk premia. In addition, increases in the ADS index are positively related to future upside premia, indicating that improved business conditions amplify compensation for positive return states, consistent with stronger speculative demand during favorable economic environments. In contrast, the downside component shows weak predictability, is limited to CRIX, and has no systematic role for sentiment or attention variables. Moreover, the absence of significant predictors for the VRP suggests that variance risk is not easily forecastable using these macro-financial and behavioral variables.

6 Conclusion

This work estimates the Bitcoin index’s return premium and variance risk premium using joint options and returns data over the most extensive available period. A Bitcoin premium decomposition is applied as a function of returns. We also propose a new functional clustering method applied to a sequence of time series of Bitcoin risk-neutral measures that allows us to obtain conditional measures for Bitcoin’s first moment and variance risk premia. Overall, we find that Bitcoin’s first and second moment risk premia, and the premium attributable to positive returns are all much larger than the corresponding measures for traditional assets like the S&P 500. We also find significant variation in these metrics between low- and high-volatility regimes, suggesting that volatility is an important state variable driving risk in the Bitcoin market. We find that

the contribution of upside return states to both the BP and BVRP is significant. Risk premia attributable to positive returns are predictable by measures of attention and sentiment and tend to be higher following improvements in business conditions.

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Appendices

A	Main Appendix	37
A.1	Data Analysis	37
A.2	Estimation of the BVIX	42
A.3	Interpolation of the IV Surface	42
A.4	Estimation of Unconditional Physical Density	44
A.5	Bitcoin Premium	45
A.6	Dimensionality Reduction and Clusters	46
A.7	Alternative Bitcoin Risk Premia Results	46
A.8	Horizon-Specific Analysis	50
A.9	Data Description: Drivers of Bitcoin Risk Premia	51
A.10	Risk Premia in the Ethereum Market	52
B	Miscellaneous	55
B.1	Further Cluster Analysis	55
B.2	Robustness: VRP	57
B.3	Cost of Carry	59
B.4	Bitcoin Comparison with Equity, Bond, and Commodity Markets	60
B.5	Related Literature on EP, VRP, and BTC options	62

A Main Appendix

A.1 Data Analysis

Two types of BTC options are traded on Deribit: those with shorter tenors—up to two days—that expire daily at 08:00 UTC and those with longer tenors that expire on Fridays at 08:00 UTC—more than two days. We utilize the data from the latter options in our study.

Figure A1 illustrates the average number of BTC option transactions per month.

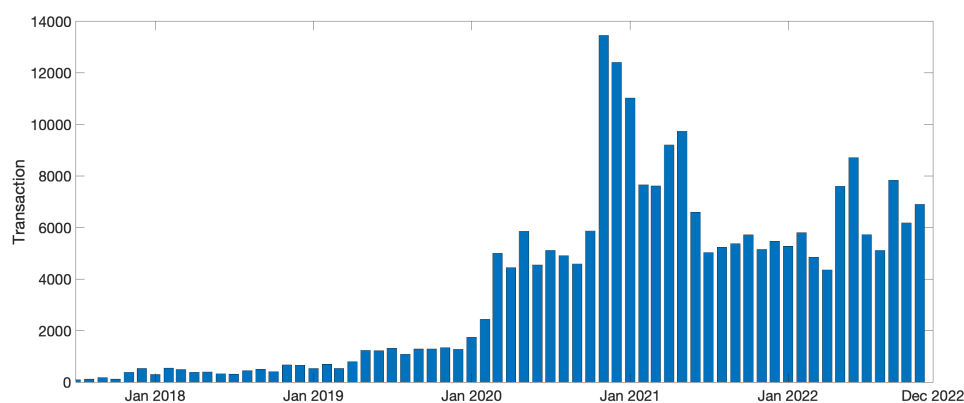


Figure A1: Average daily BTC option transaction per month

Table A2 gives an overview of the average implied volatility on different batches. We notice for both call and put options, IV initially decreases as moneyness increases and then rises past ATM, representing a "volatility smile" commonly seen in the traditional security markets. Furthermore, options with shorter maturity, particularly those deep OTM and deep ITM, tend to exhibit higher levels of IV. Notably, put options generally display higher IV compared to call options.

Table A1: Summary statistics of Bitcoin options

	Call			Put		
	TTM	Moneyness	IV	TTM	Moneyness	IV
Mean	29.27	1.21	0.82	24.57	0.91	0.89
Median	9	1.06	0.77	8	0.94	0.82
Std. Dev.	49.95	0.55	0.29	44.08	0.19	0.37
Min	1	0.08	0.05	1	0.07	0.1
Max	372	17.71	5	372	15.67	5

The table gives a summary statistic of the filtered Bitcoin call and put options traded daily from July 1, 2017, to December 17, 2022. It showcases the option characteristics, such as the time to maturity (TTM), moneyness, and the implied volatility (IV) from Deribit. The number of transactions for call options amounts to 3,940,541 and 3,468,020 for put options. Consequently, our dataset comprises 1,301 days that include a total of 7,832,590 Bitcoin option transactions, with a daily average transaction volume of 3,721 option contracts.

The summary statistic presented in Table A1 highlights the essential option characteristics, including time-to-maturity (TTM), moneyness, and implied volatility (IV) from Deribit. Similar to Teng and Härdle (2022), we find that the range of moneyness in the Bitcoin options market is significantly wider than that of traditional options markets, which can be attributed to the highly volatile nature of Bitcoin.²⁹ The average Bitcoin IV level of 82% is much higher than the average S&P 500 IV level. Furthermore, options with shorter tenors are more frequently traded than those with longer tenors. Before 2020, the average daily transaction volume was approximately 646 contracts; after 2020, this number increased to 3,721 contracts, which is almost a 476% increase.³⁰ More information about the data can be found in Appendix A.1.

²⁹Liu and Tsyvinski (2021) find that Bitcoin returns exhibit significantly higher volatility compared to stocks, ranging from 5 to 10 times greater depending on the investment horizon.

³⁰A similar trend for average daily transactions is observed in SPX options, albeit the increase is only around 20 %, from 921,948 contracts before 2020 to 1,109,514 contracts after 2020. We display the average daily Bitcoin options in Figure A1 in the Appendix.

Table A2: Implied volatility of BTC options [in level]

Moneyness	Call options				
	(0, 9]	[10, 26]	[27, 33]	>33	Average
< -0.6	4.74 (120)	2.70 (220)	1.76 (130)	1.26 (3,647)	1.45 (4,117)
[-0.6, -0.2)	1.82 (7,474)	1.26 (7,546)	1.03 (2,850)	0.95 (29,628)	1.14 (47,498)
[-0.2, 0.2]	0.78 (1,914,619)	0.75 (593,931)	0.75 (91,586)	0.78 (328,960)	0.77 (2,929,096)
(0.2, 0.6]	1.23 (87,258)	0.94 (192,329)	0.86 (49,814)	0.83 (291,264)	0.92 (620,665)
> 0.6	1.99 (4,977)	1.38 (26,024)	1.17 (13,965)	0.98 (294,199)	1.04 (339,165)
Average	0.81 (2,014,448)	0.82 (820,050)	0.83 (158,345)	0.86 (947,698)	0.82 (3,940,541)
Moneyness	Put options				
	(0, 9]	[10, 26]	[27, 33]	>33	Average
< -0.6	3.70 (69)	2.13 (1,554)	1.85 (1,429)	1.29 (34,658)	1.35 (37,710)
[-0.6, -0.2)	1.54 (97,793)	1.18 (172,290)	1.05 (39,570)	0.95 (277,739)	1.12 (587,392)
[-0.2, 0.2]	0.84 (1,776,761)	0.80 (568,189)	0.77 (82,503)	0.81 (352,108)	0.83 (2,779,561)
(0.2, 0.6]	2.31 (12,187)	1.26 (11,404)	0.81 (2,665)	0.93 (22,298)	1.35 (48,554)
> 0.6	3.15 (922)	1.92 (2,642)	1.40 (904)	1.05 (10,335)	1.36 (14,803)
Average	0.89 (1,887,732)	0.90 (756,079)	0.88 (127,071)	0.90 (997,138)	0.89 (3,468,020)

Average implied volatility of Bitcoin options across moneyness and maturity. The sample spans July 1, 2017, to December 17, 2022. Columns correspond to time-to-maturity (in days). Implied volatilities are obtained from Deribit. The average number of implied volatility observations is reported in parentheses.

Table A3: Summary statistics on option contracts of BTC options [in %]

Moneyness	Call options				Subtotal
	(0, 9]	[10, 26]	[27, 33]	>33	
< -0.6	0.00	0.01	0.00	0.09	0.10
[-0.6, -0.2)	0.19	0.19	0.07	0.75	1.21
[-0.2, 0.2]	48.59	15.07	2.32	8.35	74.33
(0.2, 0.6]	2.21	4.88	1.26	7.39	15.75
> 0.6	0.13	0.66	0.35	7.47	8.61
Total	51.12	20.81	4.02	24.05	100.00
	Put options				
	(0, 9]	[10, 26]	[27, 33]	>33	
< -0.6)	0.02	0.04	0.04	1.00	1.09
[-0.6, -0.2)	2.82	4.97	1.14	8.01	16.94
[-0.2, 0.2]	51.23	16.38	2.38	10.15	80.15
(0.2, 0.6]	0.35	0.33	0.08	0.64	1.40
> 0.6	0.03	0.08	0.03	0.30	0.43
Total	54.43	21.80	3.66	20.10	100.00

This table presents the proportion of traded BTC option contracts over moneyness and maturity. The sample covers transactions between July 1, 2017 and December 17, 2022. The columns are categorized based on the time to maturity in days. The transactions are measured as the number of traded contracts.

Table A4 provides summary statistics on option quantity in BTC units, given that each option is denominated in BTC. The distribution of quantity closely mirrors that of transaction contracts, with an even greater proportion of out-of-the-money options. Table A5 presents the summary statistics on option transaction volume in USD, calculated as the traded quantity multiplied by the option price in USD. Options with longer maturities and in-the-money options typically possess higher prices, resulting in over half of the total volume being attributed to long-maturity options. Additionally, the OTM volume portion is lower than transaction and quantity due to their lower prices.

Table A4: Summary statistics on BTC option volume [in %]

Moneyness	Call options				Subtotal
	(0, 9]	[10, 26]	[27, 33]	>33	
< -0.6	0.00	0.00	0.00	0.07	0.07
[-0.6, -0.2)	0.13	0.14	0.03	0.47	0.77
[-0.2, 0.2]	41.35	17.09	2.81	8.14	69.40
(0.2, 0.6]	2.56	5.80	1.82	9.18	19.36
> 0.6	0.15	0.95	0.41	8.89	10.40
Total	44.19	23.98	5.07	26.75	100.00

Moneyness	Put options				Subtotal
	(0, 9]	[10, 26]	[27, 33]	>33	
< -0.6)	0.00	0.04	0.03	0.84	0.91
[-0.6, -0.2)	4.09	5.83	1.50	8.85	20.28
[-0.2, 0.2]	46.47	19.21	2.82	9.28	77.78
(0.2, 0.6]	0.15	0.24	0.04	0.39	0.82
> 0.6	0.02	0.03	0.00	0.16	0.22
Total	50.74	25.35	4.40	19.51	100.00

This table presents the proportion of volume [in %] of the BTC option data over moneyness and maturity. The data spans from July 1, 2017, to December 17, 2022. The columns are categorized based on the time to maturity in days. The volume is measured in terms of the number of BTC units.

Table A5: Summary statistics on BTC option transaction value valued in USD [in %]

Moneyness	Call options				Subtotal
	(0, 9]	[10, 26]	[27, 33]	>33	
< -0.6	0.03	0.06	0.03	1.77	1.89
[-0.6, -0.2)	0.88	1.54	0.24	4.88	7.54
[-0.2, 0.2]	17.57	18.90	4.31	23.69	64.48
(0.2, 0.6]	0.26	2.07	0.95	14.14	17.41
> 0.6	0.01	0.24	0.07	8.35	8.67
Total	18.74	22.81	5.61	52.84	100.00

Moneyness	Put options				Subtotal
	(0, 9]	[10, 26]	[27, 33]	>33	
< -0.6)	0.00	0.00	0.00	0.17	0.18
[-0.6, -0.2)	0.38	1.38	0.48	8.34	10.57
[-0.2, 0.2]	20.08	19.36	3.74	27.44	70.62
(0.2, 0.6]	1.09	1.76	0.37	4.06	7.27
> 0.6	0.46	0.48	0.09	10.32	11.36
Total	22.01	22.98	4.67	50.34	100.00

This table presents summary statistics for the transaction value of BTC options. The value is measured in USD, i.e., value = volume $\times$ option price (USD) summed in each category. The data spans from July 1, 2017, to December 17, 2022. The columns are categorized based on the time to maturity in days. Within each moneyness and maturity category, the entries provide the value proportions in percentage.

A.2 Estimation of the BVIX

The BVIX is calculated using the BTC transaction data as described in Section 2. We calculate the variances σ_1^2 and σ_2^2 by closely following the original VIX methodology of CBOE and interpolate the time-weighted average as

$$\text{BVIX}_\tau = 100 \times \sqrt{\left\{ N_{T_1} \sigma_1^2 \left[\frac{N_{T_2} - \frac{\tau}{365}}{N_{T_2} - N_{T_1}} \right] + N_{T_2} \sigma_2^2 \left[\frac{\frac{\tau}{365} - N_{T_1}}{N_{T_2} - N_{T_1}} \right] \right\} \times \frac{365}{\tau}}, \quad (8)$$

where $N_{T_1} = \frac{T_1}{365}$ and $N_{T_2} = \frac{T_2}{365}$ is the time to settlement (in *years*) of the near and next-term options, respectively. A comparison of the BVIX to the Dvol index by Deribit is conducted in Figure A2, which indicates that both indices are closely related.

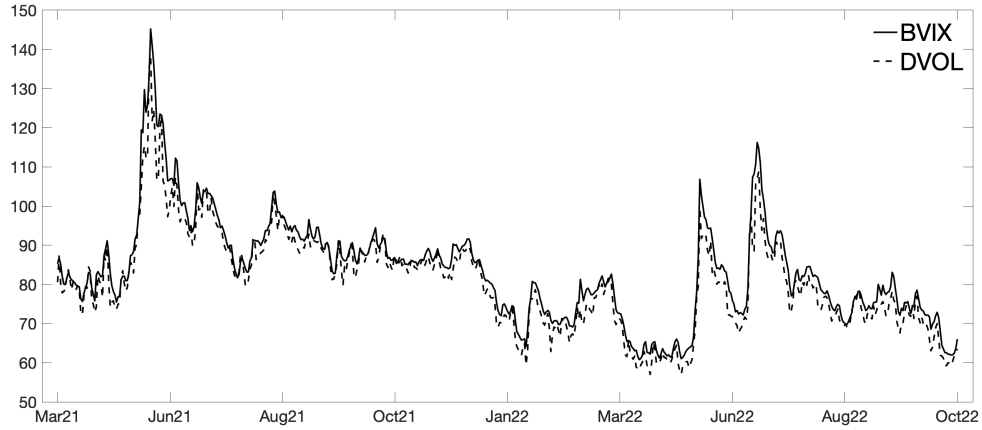


Figure A2: BVIX vs. Deribit Dvol Index

A.3 Interpolation of the IV Surface

The SVI model is widely popularized due to its parametric specification as well as good performance in the interpolation of IVs. Additionally, unlike some research assuming linearity in τ , relaxing these assumptions allows a more flexible representation of the implied volatility surface, providing a better fit to the data.

To estimate smooth surfaces, we aggregate transaction-level data for both put and call options

observed daily, using an unbalanced design that varies in terms of strike prices and time to maturity. Each day, the option data is organized as a series of strings, with each string representing a different time to maturity. Options within a string have a unique time to maturity but feature different strike prices (or levels of moneyness). Observed maturities fluctuate daily.

Following Almeida, Freire, and Hizmeri (2024), we utilize options-implied volatilities of the same maturity to fit the parametric Stochastic Volatility Inspired (SVI) model proposed by Gatheral (2004) on a daily basis. The SVI model is relatively flexible in replicating various shapes of the implied volatility curve. The implied variance can be expressed as follows:

$$\omega_{\theta}(r) = a + b \left[\rho \{r - m\} + \sqrt{\{r - m\}^2 + \sigma^2} \right],$$

where r denotes the simple return (i.e., $K/S_t - 1 \mid K = S_{t+\tau}$), and $\theta = [\alpha, \beta, \rho, m, \sigma]$ are parameters that capture various characteristics of the volatility smile, such as its level, slope, and curvature. Similarly to Gatheral (2004), to enforce no-arbitrage we impose the constraints that $b > 0$, $1 - |\rho| > 0$, $a + b \cdot \sigma \sqrt{1 - \rho^2} > 0$, and $\sigma > 0$. We estimate the parameter vector θ by minimizing the root mean squared error (RMSE)

$$\hat{\theta}_{t|\tau} = \arg \min_{\theta} \sqrt{\frac{1}{N_{t,\tau}} \sum_{i=1}^{N_{t,\tau}} \{\omega_{t,i} - \omega_{\theta}(r_{t,i}|\tau)\}^2},$$

where $\omega_{t,i}$ is the squared observed IV at day t corresponding to an option with maturity τ and strike indexed by $i \in \{1, \dots, N_{t,\tau}\}$.

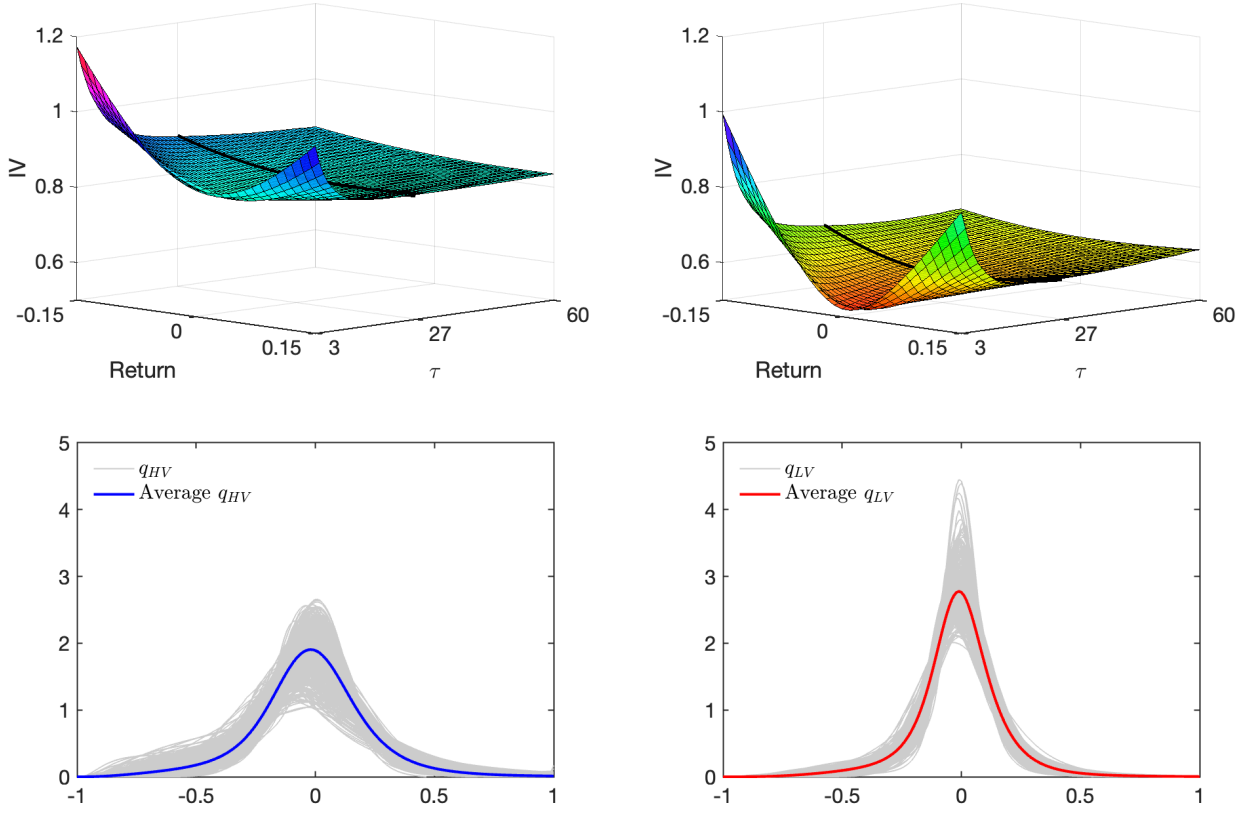


Figure A3: First row: Average IV surface interpolated by SVI for the HV cluster (left) and LV cluster (right). The black curve within each panel is the IV for TTM 27 days. Second row: Risk-neutral densities for the HV (left) and LV (right) cluster. The solid curve is the average risk-neutral density for the respective cluster.

A.4 Estimation of Unconditional Physical Density

The tails of the physical densities are estimated using the generalized extreme value (GEV) distribution following Figlewski (2008). The GEV distribution function is defined by

$$F_{GEV}(x) = \exp \left\{ - \left(1 + \xi \frac{x-a}{b} \right)^{-\frac{1}{\xi}} \right\}.$$

Specifically, the discrete empirical density is obtained via a histogram, and then a 10th-order polynomial is used to obtain the continuous density for the density between quantiles 10 and 90. The GEV distribution is used to fit the tails of the left and right parts that are below the 10th quantile and above the 90th quantile. For each tail, three parameters need to be estimated.

This requires two points to construct a loss function by making the Probability Density Function (PDF) and Cumulative Distribution Function (CDF) values at these two points close to the values of the empirical density. After all, the estimated density must integrate to one. Thus, the final estimator is

$$\hat{p}(r) = \begin{cases} \hat{f}_l^{GEV}(r), & r \leq r_{0.1}, \\ \hat{f}_{pol|hist}(r), & r_{0.1} < r < r_{0.9}, \\ \hat{f}_r^{GEV}(r), & r \geq r_{0.9}, \end{cases} \quad (9)$$

where the smoothed version of the histogram estimator is denoted by $\hat{f}_{pol|hist}$ ³¹, $\hat{f}_l^{GEV}(r)$ and $\hat{f}_r^{GEV}(r)$ are the left and right tails estimated by the GEV distribution, respectively. The 10th and 90th percentiles are denoted as $r_{0.1}$ and $r_{0.9}$, respectively.³²

A.5 Bitcoin Premium

In the estimation of empirical PDF, as $\mathbb{P}$ density, the selection of the smoothing parameter, i.e., the number of equally distant bins, can influence the shape of $\mathbb{P}$. Consequently, this affects the shape of BP. To demonstrate the robustness of BP across various smoothing parameters, Figure A4 shows the robustness of BP with different numbers of bins (NB). In the main text of the paper, we use NB of 11. Despite variations in NB, the basic shape of BP remains relatively consistent.

³¹Choosing between 8 and 13 histogram bins has a negligible impact on estimate when the number of bins within 8-13, and we use 12 bins.

³²Additionally, robustness evaluations using kernel density estimation yield results that are largely consistent with those derived from the empirical PDF, and are available upon request.

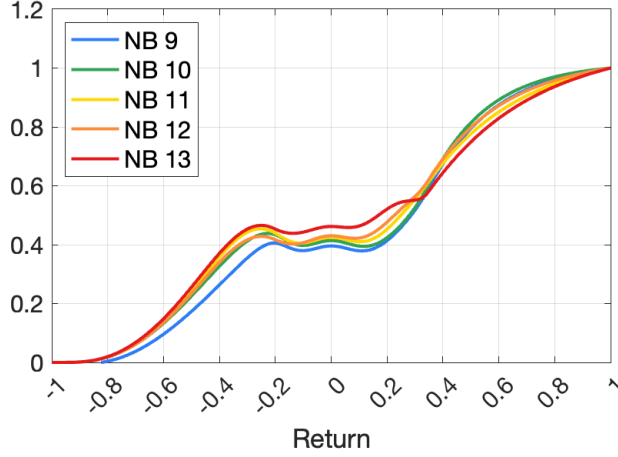


Figure A4: BP across different number of bins (NB) using empirical PDF for the $\mathbb{P}$ density with TTM 27 days.

A.6 Dimensionality Reduction and Clusters

The UMAP (Uniform Manifold Approximation and Projection) is a nonlinear dimensionality reduction technique, recently proposed by McInnes et al. (2018). It builds a topological representation of the high dimensional data set and then minimizes the following cross entropy loss function

$$\sum_{e \in E} w_h(e) \log \left(\frac{w_h(e)}{w_l(e)} \right) + (1 - w_h(e)) \log \left(\frac{1 - w_h(e)}{1 - w_l(e)} \right), \quad (10)$$

where $w_h(e)$ is the weight of the 1-simplex e in the high dimensional case and $w_l(e)$ is the weight of e in the low dimensional case. The set of all possible 1-simplices is represented as E . It has been designed to preserve the local as well as the global structure of the data. The result is illustrated in Figure A5b. Further, we illustrate the first two principal components of the distance matrix in Figure A5a.

A.7 Alternative Bitcoin Risk Premia Results

In Table A6 below, we illustrate the cluster-specific conditional estimates that rely on the time series estimates plotted in Figure A6.

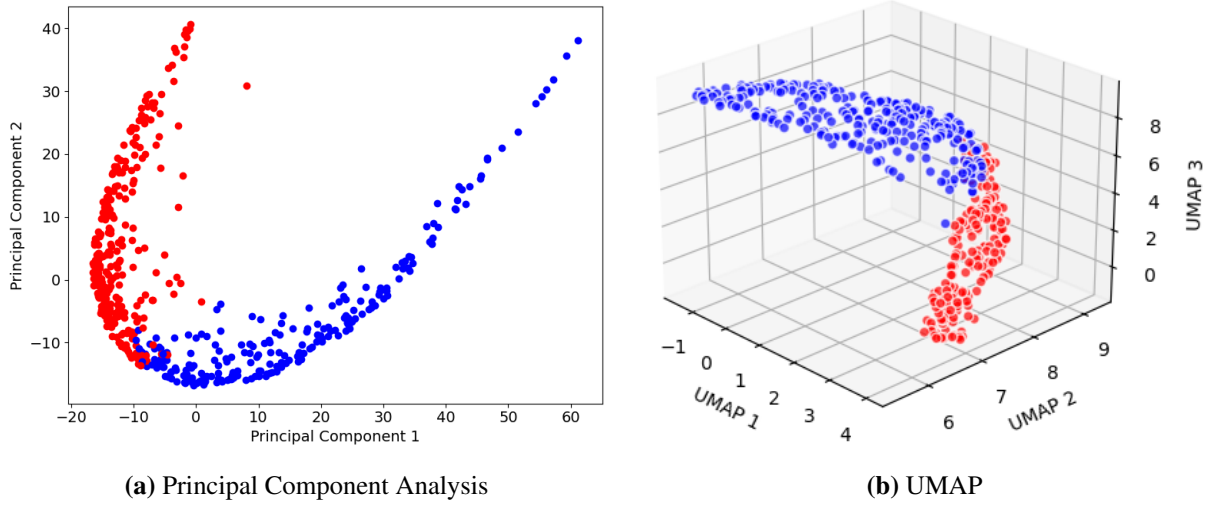


Figure A5: (a) First two principal components of the Euclidean distance matrix of risk-neutral densities. (b) Three dimensional UMAP of risk-neutral densities. Blue is the HV cluster and red is the LV cluster. Both figures refer to the *multivariate* risk-neutral density composition.

Table A6: Risk Premia Bounds: Martin (2015) and CL (2020)

Panel A: Martin (2015) – Bitcoin Premium			
	Overall	HV	LV
$\widehat{BP}^{LB}$	0.656	0.782	0.440
Panel B: CL (2020) – Bounds on Risk Premia			
	Overall	HV	LV
$\widehat{BP}^U$	0.800	0.929	0.578
$\widehat{VRP}^U$	0.147	0.180	0.090
$\widehat{BP}^R$	0.848	1.023	0.545
$\widehat{VRP}^R$	0.095	0.110	0.068

Panel A: Estimates of the lower bound on the Bitcoin risk premium from Martin (2015). **Panel B:** Unrestricted lower (U) and restricted upper (R) bounds on the Bitcoin risk premium and variance risk premium following CL (2020). All values are reported in absolute terms. Estimates are shown for the overall sample and for high- and low-volatility (HV/LV) regimes.

In Figure A6, we display the BP and VRP estimates over time that rely on Chabi-Yo and Loudis (2020), Martin (2017), and Schreindorfer and Sichert (2025). At their peak, the BP exceeded 250%, indicating extremely high required compensation for holding Bitcoin. These correspond to significant changes in Bitcoin’s price during the early 2018 crash, the March 2020 COVID-19 crash, and the 2021 regulatory news and macro events. During calm periods, e.g.,

2019, the lower bounds fall to around 50-100%, which is still higher than traditional assets. BVRP estimates are also high, particularly for Schreindorfer and Sichert (2025), and exhibit an alternating sign.

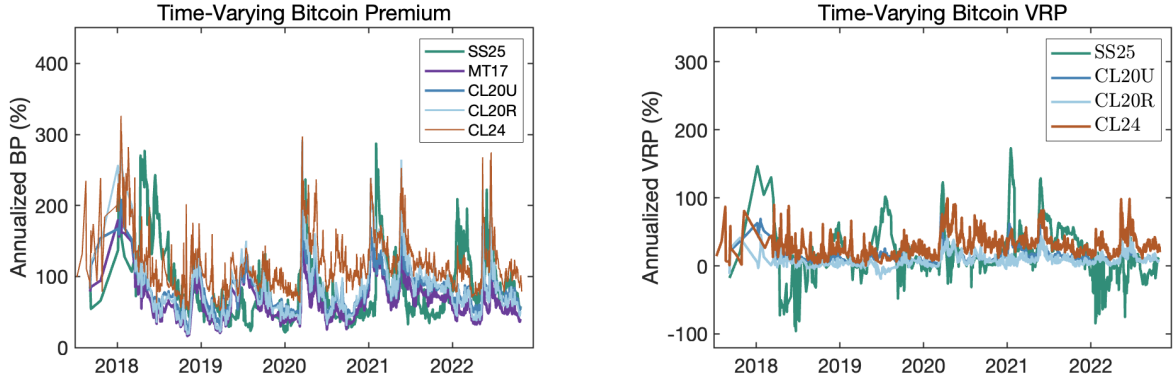


Figure A6: Time-varying Bitcoin risk premia. The left panel reports the BP, and the right panel reports the BVRP, both estimated using alternative methodologies from Martin (2017) (ML17), Chabi-Yo and Loudis (2020) (CL20) - restricted and unrestricted, Chabi-Yo and Loudis (2024) (CL24) and Schreindorfer and Sichert (2025) (SS25).

A.7.1 Schreindorfer and Sichert (2025)

This section replicates the main results in Section 5 using the estimation approach of Schreindorfer and Sichert (2025); see Tables A7 and A8, and Figure A7.

Table A7: Risk Premia (Schreindorfer and Sichert, 2025)

Panel A: Bitcoin Premium			
	Overall	HV	LV
$\widehat{BP} = \widehat{\mu}_P - \widehat{\mu}_Q$	0.59	0.68	0.45
$\widehat{\mu}_P$	0.60	0.69	0.46
$\widehat{\mu}_Q$	0.01	0.01	0.01
Panel B: Bitcoin Variance Risk Premium			
	Overall	HV	LV
$\widehat{BVRP} = \widehat{\sigma}_Q^2 - \widehat{\sigma}_P^2$	0.17	0.23	0.07
$\widehat{\sigma}_Q^2$	0.72	0.86***	0.46***
$\widehat{\sigma}_P^2$	0.54	0.63***	0.39***
Days	1017	649	368

All the estimates are annualized.

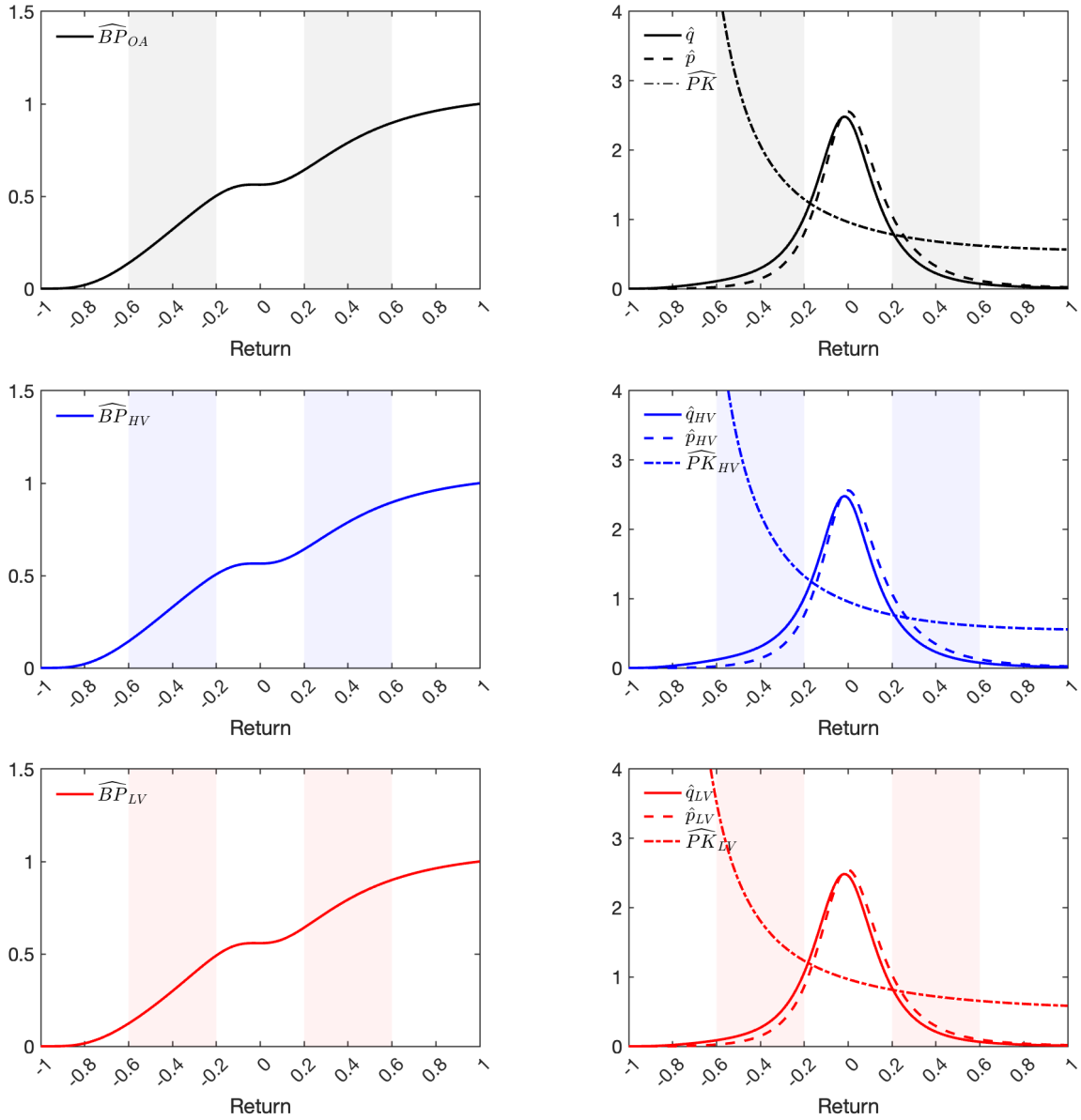


Figure A7: Results following the estimation approach of Schreindorfer and Sichert (2025): First column includes estimated BP for overall (black), HV cluster (blue), and the LV cluster (red). The second column includes estimated PK, physical, and risk-neutral density for overall (black), HV cluster (blue) and the LV cluster (red). The shaded areas mark the returns range $[-0.6, -0.2]$ and $[0.2, 0.6]$. **First row.** For the overall sample, the shaded areas contribute 35.71% and 25.13% to the overall BP, respectively. **Second row.** For the HV regime, the shaded area contributes 35.60% and 25.02% to the BP. **Third row.** The shaded areas of the LV cluster contribute 35.98% and 25.40% to the BP.

Table A8: Characteristics of BP, $\mathbb{Q}$ and $\mathbb{P}$ in influential states (Schreindorfer and Sichert, 2025)

	Negative states		Positive states		
	$\text{BP}(-0.2) - \text{BP}(-0.6)$	$\frac{\int_{-0.6}^{-0.2} q(r) dr}{\int_{-0.6}^{-0.2} p(r) dr}$	$\text{BP}(0.2) - \text{BP}(0.6)$	$\int_{0.2}^{0.6} p(r) dr$	$\frac{\int_{0.2}^{0.6} q(r) dr}{\int_{0.2}^{0.6} p(r) dr}$
Overall	0.357	0.082	0.251	0.162	0.717
HV	0.356	0.136	0.250	0.113	1.042
LV	0.360	0.087	0.254	0.150	0.755

$\text{BP}(-0.2) - \text{BP}(-0.6)$ and $\text{BP}(0.2) - \text{BP}(0.6)$ denote BP contributions over the specified intervals.
 $\int p(r) dr$ is the physical probability, and $\frac{\int q(r) dr}{\int p(r) dr}$ is the corresponding risk price.

A.8 Horizon-Specific Analysis

In this section, we extended the risk premia analysis to include 9 and 45-day horizons and investigated the robustness of our findings with respect to the volatility regime. This also enables us to characterize the term structure of the BP and BVRP.

Table A9: Bitcoin risk premia across maturities

	Overall	HV	LV
Panel A: 9-day maturity			
$\widehat{\text{BP}} = \widehat{\mu}_{\mathbb{P}} - \widehat{\mu}_{\mathbb{Q}}$	0.57	0.60	0.52
$\widehat{\mu}_{\mathbb{P}}$	0.58	0.61	0.53
$\widehat{\mu}_{\mathbb{Q}}$	0.01	0.01	0.01
$\widehat{\text{VRP}} = \widehat{\sigma}_{\mathbb{Q}}^2 - \widehat{\sigma}_{\mathbb{P}}^2$	0.20	0.30	0.02
$\widehat{\sigma}_{\mathbb{Q}}^2$	0.69	0.85	0.41
$\widehat{\sigma}_{\mathbb{P}}^2$	0.50	0.55	0.39
Days	1052	664	388
Panel B: 45-day maturity			
$\widehat{\text{BP}} = \widehat{\mu}_{\mathbb{P}} - \widehat{\mu}_{\mathbb{Q}}$	0.35	0.40	0.26
$\widehat{\mu}_{\mathbb{P}}$	0.36	0.41	0.27
$\widehat{\mu}_{\mathbb{Q}}$	0.01	0.01	0.01
$\widehat{\text{VRP}} = \widehat{\sigma}_{\mathbb{Q}}^2 - \widehat{\sigma}_{\mathbb{P}}^2$	0.41	0.51	0.23
$\widehat{\sigma}_{\mathbb{Q}}^2$	0.71	0.85	0.47
$\widehat{\sigma}_{\mathbb{P}}^2$	0.30	0.34	0.24
Days	1004	641	363

Estimates are constructed using the same methodology as in Table 2. All the estimates are annualized.

We follow the estimation approach described in Section 4. Table A9 shows that the main results for the conditional and unconditional BP and BVRP are robust across option maturities. The Bitcoin premium remains positive and economically large at all horizons, and is consistently higher in the high-volatility regime than in the low-volatility regime. Similarly, the variance risk

premium is positive across maturities and systematically larger during high-volatility periods. Across maturities, the BP tends to decline with horizon, while the BVRP increases. This pattern suggests that short-term options embed stronger speculative return premia (Gârleanu, Pedersen, and Poteshman, 2009), whereas longer maturities primarily reflect compensation for variance risk (Bollerslev and Todorov, 2011). The results for the BP contrast with the findings of Foley et al. (2022), who report an upward-sloping term structure of the Bitcoin return premium overall, with a downward-sloping pattern during Bitcoin bear markets, although their analysis considers maturities of up to 180 days.

A.9 Data Description: Drivers of Bitcoin Risk Premia

We construct a comprehensive set of predictors for Bitcoin risk premia and their components, following Chabi-Yo and Loudis (2024). These variables span four broad categories: (i) macroeconomic conditions, general uncertainty, and crypto-market volatility; (ii) Bitcoin investor sentiment; (iii) momentum and investor attention; and (iv) trading activity and option-implied trading signals. We describe the data in more detail in Section in the Appendix.

We construct a comprehensive set of predictors for the Bitcoin risk premia and their components derived using Chabi-Yo and Loudis (2024) estimation to capture the main economic, behavioral, and market-based drivers of Bitcoin risk premia and their components. These variables span four broad categories.

Macroeconomic conditions, general uncertainty and crypto-market volatility. We include the Economic Policy Uncertainty (EPU) index of Baker, Bloom, and Davis (2016), which measures uncertainty related to fiscal, monetary, and regulatory policies at a monthly frequency from www.policyuncertainty.com. In addition, we use the Aruoba-Diebold-Scotti (ADS) business conditions index of Aruoba, Diebold, and Scotti (2009) from the Federal Reserve Bank of Philadelphia, a daily indicator that summarizes real-time economic activity. To measure uncertainty specific to cryptocurrency markets, we include the CRyptocurrency IndeX (CRIX) of Trimborn and Härdle (2018) from <https://www.royalton-crix.com>, which aggregates volatility across major cryptocurrencies and serves as a proxy for overall market conditions in the crypto ecosystem.

The latter data set is available only from the start of 2020.

Bitcoin investor sentiment. We incorporate measures of investor sentiment obtained from the Sentix database available via Bloomberg. This dataset is weekly and starts on 08.09.2017. We extract five indices: Sentiment Headline Index, Overconfidence Index, Institutional Investors, Individual Investors, Bullish, and Bearish. These data are based on a large-scale survey of global investors, including both private and institutional participants, and reflect expectations over short horizons (typically one month). The Headline Sentiment Index is constructed as the net balance between bullish and bearish responses, providing a forward-looking measure of market expectations, and was used in (Anamika, Chakraborty, and Subramaniam, 2023) to explain Bitcoin price movements.

Momentum and investor attention. Following Liu and Tsyvinski (2021), we proxy investor attention using Google Trends search intensity for the term “Bitcoin,” which reflects retail interest and information demand. Momentum is constructed as the average return over the previous week.

Trading activity and option-implied trading signals. Trading activity on the Bitcoin options market is proxied by (i) total trading volume and (ii) value-weighted volume of call and put options on each day t , capturing liquidity and market participation. To extract directional signals, we compute the put–call ratio,

$$X_{it} = \frac{P_{it}}{P_{it} + C_{it}},$$

where P_{it} and C_{it} denote the trading volume of put and call options, respectively, following Pan and Poteshman (2006).

A.10 Risk Premia in the Ethereum Market

Daily USD-denominated Ethereum prices are collected from CoinGecko from August 2015 to March 2026. We use daily transaction-level options data from Deribit spanning March 2019 to December 2025. We follow the same data-cleaning procedure as illustrated in Section 2 for Bitcoin, and the estimation methodology outlined in Section 4.2. Several noteworthy features emerge in the analysis of the Ethereum market.

First, Table A10 summarizes the results for the unconditional and conditional risk premia. The Ethereum return premium (ETHP) is high, nearly double that of the Bitcoin premium reported in Table 2. In contrast, the Ethereum variance risk premium (ETHVRP), while still high compared to the index market, is less than half the premium we find for Bitcoin.

Second, Table A11, and Figure A8 illustrate how risk premia are allocated over return states. More than 60% of the ETHP is attributable to positive returns between 20% and 60%, independently of the market regime. At the same time, the ETHP attributable to negative returns between -60% and -20% is less than 5%.

These results suggest that the Ethereum market is largely driven by speculative agents who are primarily concerned with upside risk across all market regimes. In addition, investors appear to place relatively less weight on the variance premium compared to the Bitcoin market. Overall, the findings are qualitatively similar across the two markets. However, further work is needed to better understand the differences between them.

Table A10: Risk Premia (Ethereum)

Panel A: Ethereum Premium			
	Overall	HV	LV
$\widehat{\text{ETHP}} = \hat{\mu}_{\mathbb{P}} - \hat{\mu}_{\mathbb{Q}}$	1.37	1.43	1.14
$\hat{\mu}_{\mathbb{P}}$	1.39	1.45	1.17
$\hat{\mu}_{\mathbb{Q}}$	0.03	0.02	0.03
Panel B: Ethereum Variance Risk Premium			
	Overall	HV	LV
$\widehat{\text{ETHVRP}} = \hat{\sigma}_{\mathbb{Q}}^2 - \hat{\sigma}_{\mathbb{P}}^2$	0.08	0.05	0.15
$\hat{\sigma}_{\mathbb{Q}}^2$	0.69	0.76	0.48
$\hat{\sigma}_{\mathbb{P}}^2$	0.61	0.71	0.32
Days	116	87	29

Panel A: Estimates of the unconditional EP and conditional i . **Panel B:** Estimates of the unconditional ETHVRP and conditional ETHVRP $_i$. Unconditional estimates are referred to as 'Overall', and the conditional ones are cluster-specific for $i \in \{\text{HV}, \text{LV}\}$. $\hat{\mu}_{\mathbb{P}} = \frac{365}{27} \int_{-1}^1 x \{\hat{p}(x)\} dx$ and $\hat{\sigma}_{\mathbb{P}}^2(R)$ is the sample mean of realized variances. $\mu_{\mathbb{Q}}$ is estimated using 1-month Treasury bill rates and for $\hat{\sigma}_{\mathbb{Q}}^2(R)$ we use square of ETHVIX. All the estimates are annualized.

Table A11: Characteristics of ETHP, $\mathbb{Q}$ and $\mathbb{P}$ in influential states (Ethereum)

	Negative states			Positive states		
	ETHP(-0.2)-ETHP(-0.6)	$\int_{-0.6}^{-0.2} p(r)dr$	$\frac{\int_{-0.6}^{-0.2} q(r)dr}{\int_{-0.6}^{-0.2} p(r)dr}$	ETHP(0.2)-ETHP(0.6)	$\int_{0.2}^{0.6} p(r)dr$	$\frac{\int_{0.2}^{0.6} q(r)dr}{\int_{0.2}^{0.6} p(r)dr}$
Overall	0.026	0.123	1.123	0.652	0.388	0.320
HV	0.043	0.125	1.192	0.614	0.384	0.344
LV	-0.022	0.109	0.926	0.785	0.393	0.249

ETHP(-0.2)-ETHP(-0.6) and ETHP(0.2)-ETHP(0.6) are ETHP contributions over the corresponding intervals. $\int p(r)dr$ denotes the physical probability over these states and $\frac{\int q(r)dr}{\int p(r)dr}$ is the associated risk price.

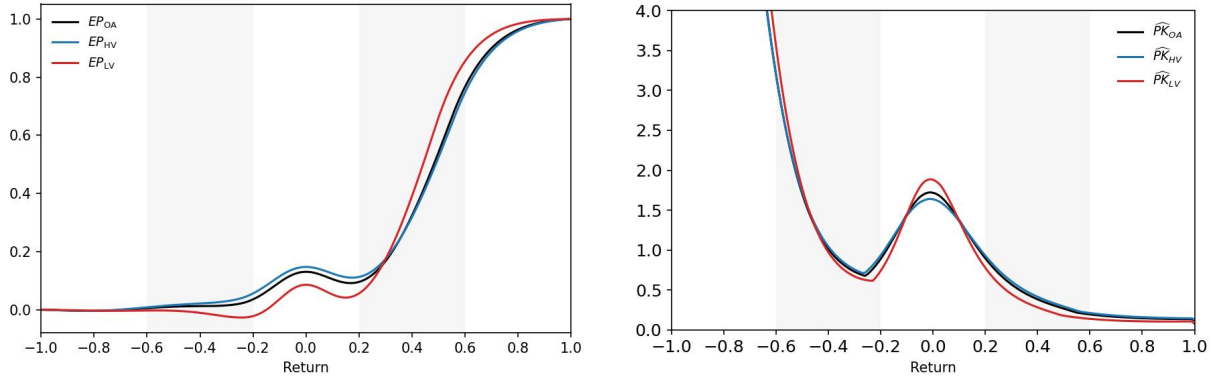


Figure A8: Estimated premium across return states. Each panel plots the unconditional (black) and conditional estimates for the high-volatility (blue) and low-volatility (red) regimes. The shaded areas indicate the return ranges $[-0.6, -0.2]$ and $[0.2, 0.6]$, which are used to assess the contribution of influential states.

B Miscellaneous

This appendix presents miscellaneous results to support our empirical arguments, particularly through further robustness checks.

B.1 Further Cluster Analysis

Figure B1 provides another way to view $\mathbb{Q}$ -density in HV and LV clusters for time-to-maturity 27 days.

Table B1 displays the logistic regression of the clusters on BP, BVIX, and VRP. It reveals that BP calculated by realized return minus risk-free rate can not explain the cluster variation. BVIX shows substantial explainable power, accounting for nearly half of the cluster variance. VRP calculated by q variance minus realized variance is not significant. The coefficients indicate that a higher BVIX index is associated with a higher probability of HV cluster, i.e., the high volatility cluster. Table B2 displays the logistic regression examining the relationship between the clusters and the first four moments of risk-neutral density. The variance explains 69% of the variation in clusters on its own. When combined, all four moments together account for 70% of the cluster variation. Table B3 presents the logistic regression of clusters with single factors using time-to-maturity 27 days. The dependent variable is the cluster label, while the independent variables include realized returns (RR), realized variance (RV), BVIX, $\mathbb{Q}$ variance, VRP(calculated either by BVIX or RV), jumps (including negative and positive jumps) and sentiment index. Significant factors include RR, RV, BVIX, $\mathbb{Q}$ variance, and negative and positive jumps separately. Only RV, BVIX, and $\mathbb{Q}$ variance have good explanatory power. Figure B2 displays the distribution of daily average BTC option transactions, categorized by clusters and also differentiated into call and put options. This visual illustration confirms that most transactions are OTM for both call and put options, aligning with the summary statistics of transactions we showed in Table A3. Regarding the clustering aspect, the figure indicates that cluster 0 typically experiences higher daily average transactions than cluster 1.

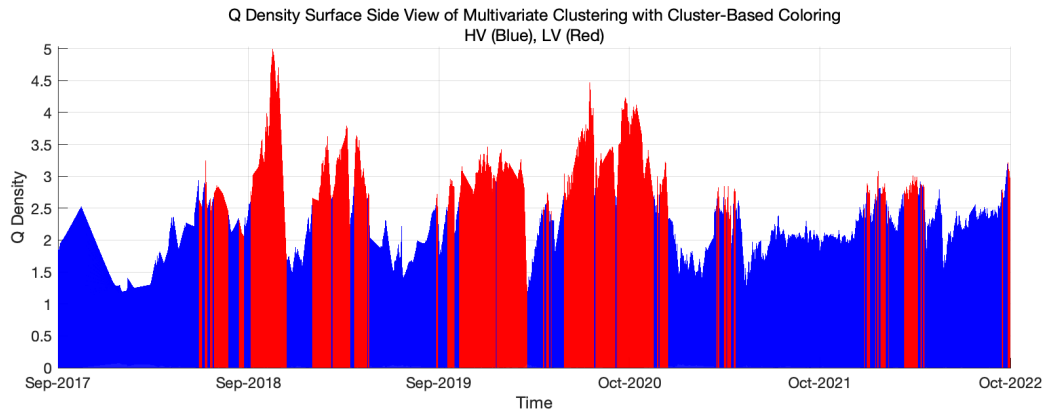


Figure B1: Time varying risk-neutral densities viewed from the side with time to maturity 27 days. The HV cluster is colored in blue and the LV cluster is colored in red.

Table B1: Logistic Regression of Clusters on BP, BVIX, and VRP

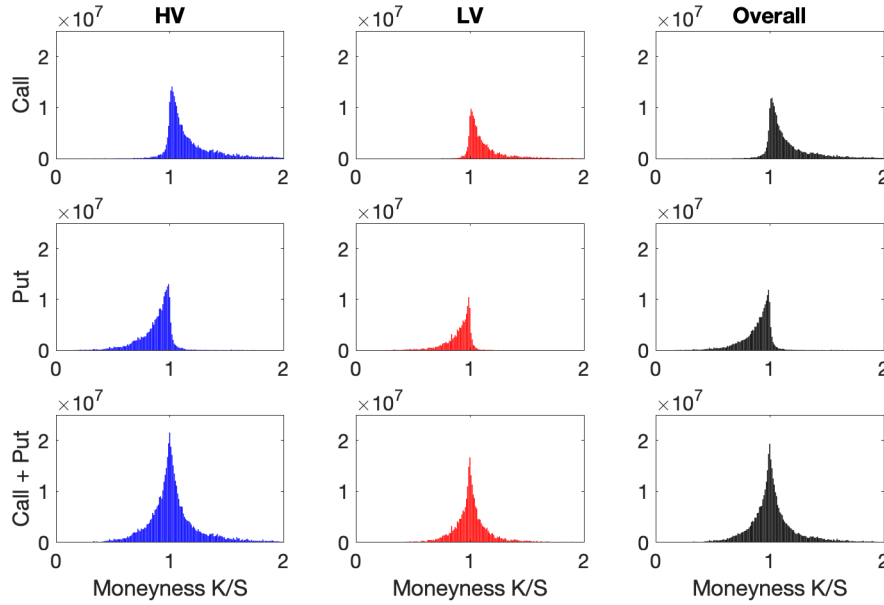
	(1)	(2)	(3)	(4)	(5)
Constant	-0.20** (0.09)	-1.21*** (0.18)	-0.25*** (0.09)	-1.21*** (0.18)	-0.25*** (0.09)
BP	0.12 (0.09)			-0.25 (0.18)	0.11*** (0.10)
BVIX		-3.75*** (0.33)		-5.37 (0.34)	
VRP			0.15 (0.11)		0.12 (0.11)
R2	0.00	0.49	0.00	0.50	0.01
Adj. R2	-0.00	0.45	0.00	0.49	0.00
T_{HV}	278	271	271	271	271
T_{LV}	227	211	211	211	211
T	505	482	482	482	482

This table displays the logistic regression of the clusters on BP, BVIX, and VRP. The dependent variable is the cluster label. The independent variables are BP, BVIX, and VRP. The number of observations in the HV cluster, LV cluster, and the overall sample is denoted as T_{HV} , T_{LV} and T , respectively. It reveals that BP calculated by realized return minus risk-free rate can not explain the cluster variation. BVIX shows substantial explainable power, accounting for nearly half of the cluster variance. VRP calculated by q variance minus realized variance is not significant. The coefficients indicate that a higher BVIX index is associated with a higher probability of HV cluster, i.e., the high volatility cluster.

Table B2: Logistic Regression of clusters on first four moments

	(1)	(2)	(3)	(4)	(5)
Constant	-0.43*** (0.11)	-2.50*** (0.33)	-0.46*** (0.11)	-0.18 (0.15)	-2.19*** (0.65)
Mean	1.55*** (0.21)				0.30 (0.40)
Variance		-7.70*** (0.80)			-6.28*** (1.78)
Skewness			1.85*** (0.20)		-0.04 (0.74)
Kurtosis				3.93*** (0.36)	1.04 (1.50)
R2	0.14	0.69	0.24	0.60	0.70

This table displays the logistic regression examining the relationship between the clusters and the first four moments of risk-neutral density. The dependent variable is the cluster label, and the independent variables include annualized mean, annualized variance, skewness, and excess kurtosis of risk-neutral density. All four moments are standardized. The number of observations in the high-volatility (HV) cluster, low-volatility (LV) cluster, and the overall sample is $T_{HV} = 278$, $T_{LV} = 227$ and $T = 505$, respectively. The variance explains 69% of the variation in clusters on its own. When combined, all four moments together account for 70% of the cluster variation.

**Figure B2:** Daily average transaction distribution for different clusters and for call and put, respectively.

B.2 Robustness: VRP

To provide more comprehensive VRPs, we provide alternatives of $\mathbb{Q}$ -variance and $\mathbb{P}$ -variance. For the robustness check of $\mathbb{P}$ -variance, we use the second moment of $\mathbb{P}$ density estimated by

Table B3: Logistic regression of clusters, with single factors TTM=27

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Constant	-0.20** (0.09)	-1.08*** (0.17)	-1.22*** (0.181)	-2.50*** (0.33)	-0.25*** (0.09)	-0.25*** (0.09)	-0.20** (0.09)	-0.24*** (0.09)	-0.23** (0.09)	-0.20*** (0.09)
RR	0.15* (0.09)									
RV		-3.90*** (0.41)								
BVIX			-3.75*** (0.33)							
Q variance				-7.70***						
VRP(BVIX)					0.15 (0.11)					
VRP(RV)						0.15 (0.11)				
Jump							0.07 (0.45)			
Jump(-)								0.60*** (0.18)		
Jump(+)									-0.48*** (0.16)	
Sentix										0.02 (0.09)
R2	0.00	0.28	0.49	0.69	0.00	0.00	0.00	0.02	0.02	0.00
Num. Cluster 0	278	271	271	278	271	271	278	278	278	278
Num. Cluster 1	227	211	211	227	211	211	227	227	227	227
Num. Total	505	482	482	505	482	482	505	505	505	505

empirical PDF and present VRP in Table B4. Figure B3 presents two measures of VRP over time; the left is based on the empirical risk-neutral variance, and the right is based on BVIX.

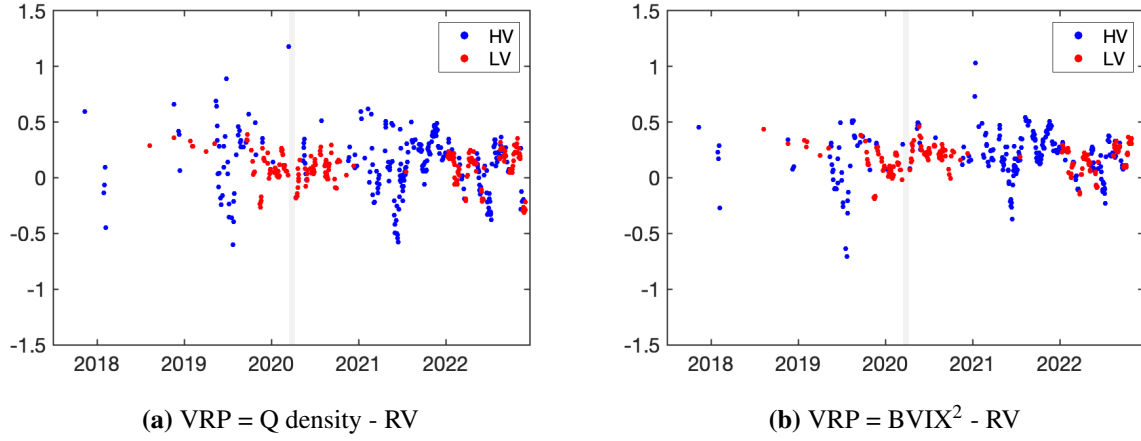


Figure B3: VRPs over time. In a) VRP is calculated by the second moment of empirical risk-neutral density minus RV. In b) VRP is calculated by BVIX^2 minus RV.

Table B4: Robustness check: Risk Premia

	Based on $\mathbb{Q}$ density			Based on BVIX		
	Overall	HV	LV	Overall	HV	LV
$\text{Var}_{\mathbb{Q}}(R)$	0.63	0.80***	0.43***	0.71	0.88***	0.50***
$\text{Var}_{\mathbb{P}}(R)$	0.68	0.71	0.32	0.68	0.71	0.32
VRP	-0.04	0.09***	0.11***	0.04	0.16***	0.18***
Observations	505	278	227	482	271	211

$\text{Var}_{\mathbb{P}}(R)$ is σ_p^2 .

B.3 Cost of Carry

Figure B4 shows the cost of carry, calculated using BTC futures data from Liu, Sepp, and Packham (2023), compared with the first moment of our estimated $\mathbb{Q}$ density.

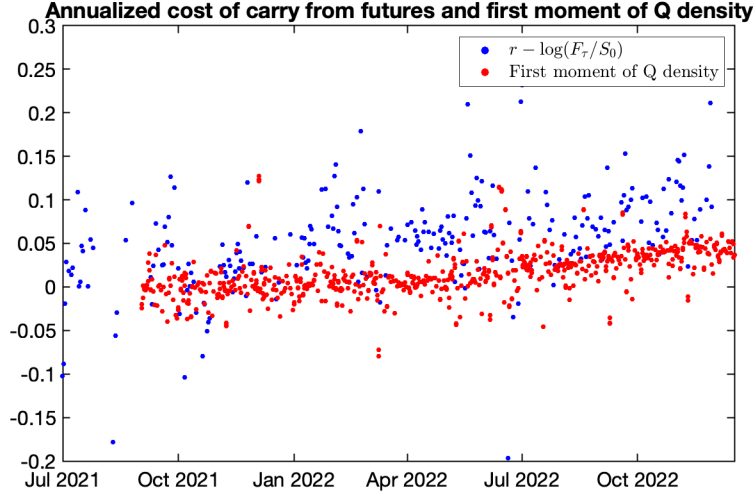


Figure B4: Cost of carry, calculated using BTC futures data from Liu, Sepp, and Packham (2023), compared with the first moment of our estimated $\mathbb{Q}$ density

B.4 Bitcoin Comparison with Equity, Bond, and Commodity Markets

Table B5 compares the SR of BTC with equity, bond, and commodity markets. It shows the SRs calculated by simple returns and log returns. Since BTC returns are asymmetric with high volatility, simple and log returns show different SRs. For simple returns, the SR of BTC is higher than that of other markets, while from log returns, it is around the same level as that of S&P 500 market.

Table B5: Sharpe Ratio (monthly and annual)

Panel A: log returns							
	Monthly			Annual			Obs.
	$\hat{\mu}$ (%)	$\hat{\sigma}$ (%)	$\widehat{SR}$	$T\hat{\mu}$ (%)	$\sqrt{T}\hat{\sigma}$ (%)	$\sqrt{T}\widehat{SR}$	
BTC	2.89	21.83	0.13	34.68	75.61	0.46	108
S&P 500	0.68	4.41	0.15	8.14	15.28	0.53	108
Russel 2000	0.39	5.81	0.07	4.64	20.12	0.23	108
US Bond	0.05	1.10	0.05	0.61	3.80	0.16	103
US Commodity	-0.06	7.06	-0.01	-0.73	24.45	-0.03	103

Panel B: simple returns							
	Monthly			Annual			Obs.
	$\hat{\mu}$ (%)	$\hat{\sigma}$ (%)	$\widehat{SR}$	$T\hat{\mu}$ (%)	$\sqrt{T}\hat{\sigma}$ (%)	$\sqrt{T}\widehat{SR}$	
BTC	5.40	23.33	0.23	64.81	80.82	0.80	108
S&P 500	0.78	4.40	0.18	9.32	15.23	0.61	108
Russel 2000	0.55	5.73	0.10	6.63	19.86	0.33	108
US Bond	0.06	1.10	0.05	0.68	3.80	0.18	103
US Commodity	0.18	6.83	0.03	2.16	23.67	0.09	103

Sharpe Ratio: $SR = (\mu - R_f)/\sigma = (E_P(R) - R_f)/\sqrt{Var_P(R)}$. We use $\hat{\mu} = \frac{1}{T} \sum_{t=1}^T R_t$, $\hat{\sigma} = \sqrt{\frac{1}{T} \sum_{t=1}^T (R_t - \hat{\mu})^2}$ to estimate Sharpe Ratio, $\widehat{SR} = (\hat{\mu} - R_f)/\hat{\sigma}$. R_t are **monthly** log returns in Panel A and simple returns in Panel B. Risk-free rate $R_f = 0$. Annualized Sharpe Ratio $\sqrt{T}\widehat{SR}$, with T -observations annually, $T = 12$ for all. These time series are from January 1, 2014 to December 31, 2022, consistent with the BTC daily prices we used in this paper. For BTC, the annualized simple return $T\hat{\mu}$ (0.64) and volatility $\sqrt{T}\hat{\sigma}$ (0.81) are consistent with the unconditional return $\hat{\mu}_{\mathbb{P}}$ (0.70) and variance $\hat{\sigma}_{\mathbb{P}}^2$ (0.56) in Table 2, the minor difference might come from the bandwidth we use to calculate P -density. US Bond and US Commodity are represented by the S&P US Treasury Bond Index and S&P GSCI Index, respectively, which have been freely available only since 2014-06-01, so they have fewer observations. The Sharpe Ratio calculation refers to Lo (2002). Compared to Chen and Vinogradov (2021), they get a BTC SR of 0.6.

Table B6: Correlation matrix

	BTC	S&P 500	Russel 2000	US Bond	Global Commodity
BTC		-0.02	-0.01	0.00	0.06***
S&P 500			0.88***	-0.22***	0.32***
Russel 2000				-0.20***	0.33***
US Bond					-0.17***
Global Commodity					

For equity markets, we use S&P 500 and Russel 2000 indices. For the bond market, we utilize the S&P US Treasury Bond Index. For Global commodities, we use the S&P GSCI Index. Correlation is performed using a t-test (H_0 : no correlation). The t-statistic is calculated as $t = \frac{corr\sqrt{n-2}}{\sqrt{1-corr^2}}$, where $corr$ represents the correlation coefficient and n is the sample size. Significance levels are denoted by 1%(***) , 5%(**) and 10%(*) . These time series are from June 6, 2014 to December 31, 2023.

Table B6 presents the correlation matrix among Bitcoin, S&P 500, Russel 2000, US Bond, and Global Commodity indices. This indicates that BTC has no significant correlation with equity and bond markets but is more correlated with commodity markets.

B.5 Related Literature on EP, VRP, and BTC options

Table B7: Stock Equity premium and Bitcoin premium

Panel A: Stock		
	Equity premium	Time
Heston, Jacobs, and Kim (2023)	0 - 8.32%	1996 - 2019
Tetlock (2023)	8.64%	1996 - 2021
Chabi-Yo and Loudis (2024)	8.72%	1996 - 2019
Panel B: BTC		
	BTC premium	Time
Chen and Vinogradov (2021)	48.12%	Feb 2018 - Sep 2020
Foley et al. (2022)	80%	2018 - 2020
Wilson (2024)	273.6%	Apr 2010 - Feb 2023
	52.68%	Dec 2013 - Feb 2023
	75%	May 2017 - Feb 2023

Table B8: Stock and Bitcoin Variance Risk premium

Panel A: Stock			
	VRP	Time	Definition
Bakshi and Kapadia (2003)	(-)	Jan 1, 1988 - Dec 31, 1995	reg. coef. b/w DH
			gains and vega
Carr and Wu (2009)	-2.74%	Jan 1996 - Feb 2003	$P - Q$
Bollerslev, Tauchen, and Zhou (2009)	18.30% ²	1990 - 2007	$Q - P$
	monthly		
Todorov (2010)	-0.4015% ²	1990 - 2002	$P - Q$
	daily		
Bekaert and Hoerova (2014)	(+)	Jan 2, 1990 - Oct 1, 2010	$Q - P$
Zhou (2018)	(+)	1990 - 2015	$Q - P$
Rombouts, Stentoft, and Violante (2020)	17%	Jan 1990 - Sep 2015	$Q - P$
Heston, Jacobs, and Kim (2023)	-6.06% - 0	1996 - 2019	λv_t
Tetlock (2023)	1.56%	1996 - 2021	$Q - P$
Panel B: BTC			
	VRP	Time	Definition
Alexander and Imeraj (2021)	mostly (-)	Mar 2019 - Mar 2020	$P - Q$

In Bollerslev, Tauchen, and Zhou (2009), the VRP of S&P 500 index is reported as 18.30%², so the annualized VRP is approximately $18.30\%^2 \times 12 = 219.6\%^2$ or 2.196%. In Todorov (2010), the VRP is reported as -0.4015 in variance unit (%²), therefore the annualized VRP is $-0.4015\%^2 \times 252 = 101.178\%^2$ or 1.012%.

Table B7 summarizes empirical results from the literature about the stock equity premium and the Bitcoin premium. Notably, Chabi-Yo and Loudis (2024) decomposed EP into bad, moderate, and good states by moneyness. They find that the average contribution in bad states is about 20%

of EP, while in crisis, this number increases to 60%. In a stable period, the central state contributes 80%. They claim they also decompose higher moments of VRP in online supplements. Another point to notice is that Wilson (2024) defines Bitcoin Premium as excess returns of BTC returns minus stock returns. Further, table B8 reports results on the literature of equity VRP and Bitcoin VRP. Bollerslev, Tauchen, and Zhou (2009) and the best model of Bekaert and Hoerova (2014) have similar tendency, as shown in Cheng (2019).

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